Rank two projective geometries

Complete source treatment and shared proofs for L063 and L064

The published Part II article and the retained appendix are separately registered originals. Their individual claim ledgers accompany this volume. The opening manuscript identifies every correction and the classical premises; the following papers contain the full proofs. The omitted affine family and all previously accepted components are reused without repeating their acquisition. Earlier scope statements inside those exact components retain their historical wording; the opening manuscript records their completed source joins. No new principal-refutation count, compression ratio or Lean verification is claimed.

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Rank two projective geometries: classification and corrections

Human Compression Project

Abstract

We give a complete treatment of the mathematical claims in the fixed published Part II article and its retained appendix. The published classification omits affine configurations over subfields. Its convention identifies the two parabolic subgroups without ordering them, which also changes two families of conjugacy counts. Four stated full graph groups require correction. The surviving configurations, their class counts, their local arc lengths and the remaining full group assertions follow from the proofs collected here. We distinguish these corrections from empty parameter cases, preliminary slips, classical premises and the contents missing from the retained appendix edition.

1 The two originals and the meaning of completion

The first original, designated L063, is Buekenhout, De Saedeleer and Leemans, *On the rank two geometries of the groups* PSL$(2, q)$*: part II*, Ars Mathematica Contemporanea 6 (2013), 365–388. It has 24 pages. The second, L064, is a retained 21-page appendix edition beginning with the proof of Lemma 12 and ending with Proposition 30. They are two separately registered originals. Shared arguments are included once in this collection, while the claim ledgers identify each original and its actual wording separately.

A complete treatment of a false assertion is its explicit disproof, not an attribution of the corrected assertion to the source. The omitted affine family was already established in the earlier accepted component; it is reused here. The present completion adds the proofs needed for the surviving assertions. It does not record another principal refutation of either original. In particular, the source's four impossible square-field rows are empty conditional cases, rather than four numerical counterexamples.

Our scope is the mathematical classification and its stated consequences. Historical computer timings and statements that particular examples were checked in MAGMA are attributed to their authors; those historical runs are not reproduced. The classification does not depend on them. The authors' earlier constructions, the Biggs–Hoare and Wong graphs, and the classical generalized quadrangle retain their original attribution.

2 A common proof structure

Put $G = \mathrm{PSL}(2, q)$, and write $\{M, L\}$ for two distinct proper parabolic subgroups, with $B = M \cap L$. We use the unordered convention in Section 2 of L063. Conjugacy means an orbit under G; isomorphism in that source means an orbit under $\mathrm{Aut}(G)$. This latter convention concerns the distinguished action and does not assert that every abstract graph isomorphism is induced by $\mathrm{Aut}(G)$.

The coset construction gives one object for each coset of M or L, with incidence defined by nonempty intersection. Every incident pair is translated to the standard pair, whose stabilizer

is B. Its local actions are M on M/B and L on L/B. Thus the required rank two conditions amount to

$$M \text{ or } L \text{ maximal in } G, \qquad |B\backslash M/B| = |B\backslash L/B| = 2.$$

The local actions are primitive and have degree at least two, proving firmness. The other parabolic is not contained in the maximal one, so the two generate G; this proves connectedness. For fixed M, ordered configurations are normalizer orbits on L. If the parabolic types agree, one must also identify the orbits obtained by exchanging them. This is the accepted common rank two argument from the Part I treatment, restated to fix the convention used throughout this collection.

The next step is to classify the possible two-transitive actions. For $d = [H : B]$, a point stabilizer is transitive on the other $d - 1$ points, so

$$d - 1 \mid |B|, \qquad |H| \leq |B|(|B| + 1).$$

The separately proved common action theorem combines this bound with the classical subgroup types, and treats the normal PSL subgroup of PGL explicitly. It supplies the projective, affine and small exceptional possibilities used in every subsequent application. We do not assume the source's representation proposition, whose cited proof was listed as being in preparation.

Proof component	Its role in the classification
Two-transitive coset actions in dimension two	Complete candidate lists for the local parabolics and intersections.
Subfield planes and unordered pairs of parabolics	Affine planes, their conjugacy and Frobenius orbits, and the odd-characteristic unordered count.
Counting configurations above a cyclic subgroup of order three	All seven Table 2 rows, including the corrected equal-type inner count.
Exceptional parabolics from their common normalizers	Tables 1 and 3 exceptional counts and the empty square-field cases.
Subfield parabolics and affine overgroups	Complete even, odd and quadratic subfield alternatives, including the small fields.
Local arc transitivity from small stabilizers	Every local arc column in Tables 1–6, including the nine Table 9 crossreferences.
The three graph group components	Full groups for the S_4, S_4 families, the S_4, D_{16} subdivisions, and the $q = 9$ quadrangle.

The subgroup classification of Dickson, Moore and Suzuki is an explicit classical premise, including the stated maximal subgroups, subfield embeddings, normalizers and fusion under PGL. The order of PSL and its action on a projective line are classical as well. The full graph group proofs additionally use Tutte's bound of 48 for a vertex stabilizer in a finite connected cubic arc-transitive graph, under its actual hypotheses. None of these premises is a retained version of the target Part II geometry classification. The accepted Part I treatment is used only where a maximal parabolic belongs to its Borel or dihedral scope.

3 The corrected classification and numerical columns

The complete action list separates the maximal parabolics of the Part II problem into A_4, S_4, A_5, proper PSL subfield groups and quadratic PGL subfield groups. For each allowed intersection,

the common normalizer proofs count actual configurations, rather than inferring their number merely from subgroup conjugacy. The exceptional and small field arguments supply all finite branches. The subfield proof shows that there are no further branches.

For clarity, the changes to the printed numerical columns are the following. All other nonempty rows retain their source class counts.

1. In even characteristic, let $K = \mathbb{F}_s \subset F = \mathbb{F}_{s^m}$ with $s \geq 4$ and m prime. Beside $M = \mathrm{PSL}(2, K)$, take

$$B = U_K : K^*, \qquad L = U_V : K^*, \qquad K \subset V \subset F, \quad \dim_K V = 2.$$

   Every such V gives the required configuration. There are $(s^{m-1} - 1)/(s - 1)$ conjugacy classes, and the automorphism classes are exactly the Frobenius orbits on these subspaces. At $m = 2$ this is the single printed Table 4 row. The assertion that $m = 2$ is necessary is false. At $s = 4$ the accepted scalar formulas recover the Table 1 counts as well.

2. In Table 2 rows 6–7, write $m = (p - \epsilon)/6$, where $3 \mid (p - \epsilon)/2$. The printed $2m - 1$ counts ordered configurations. For unordered configurations the correct inner count is

$$m + \lfloor m/2 \rfloor.$$

   The automorphism count remains m. The proof identifies the overgroups with a regular dihedral normalizer quotient and then takes inversion orbits.

3. In Table 5, for $q = 3^m$ with m an odd prime, the inner count is $(3^{m-1} - 1)/2$, rather than $3^{m-1} - 1$. At $q = 27$ it is four rather than eight. The printed automorphism counts are retained: two at $m = 3$, and $(3^{m-1} - 1)/(2m)$ otherwise.

4. In the square-field part of Table 1, $q = p^2 \equiv -1 \pmod 5$ with p odd implies $q \equiv 9 \pmod{40}$. Rows 12–14 and 18 therefore have no admissible parameters. Rows 15–17 remain; the alternative $q \equiv -1 \pmod{40}$ in rows 16–17 is also empty.

The complete source tables have $18 + 7 + 9 + 5 + 2 + 3 = 44$ rows. This is a count of printed rows, not of distinct families or new theorems. Table 4 row 5 repeats Table 1 row 1; the $s = 4$ subfield case overlaps the A_5 discussion; four printed square-field rows are empty. The ledgers record these overlaps without adding source credit. The nine entries of Table 9 point back to those same rows.

The original local arc lengths are proved for each actual configuration. The added even-characteristic affine configurations have local arc length three by the same local calculation. The S_4, S_4, D_8 configuration has length four for the specified simple group, and the S_4, D_{16}, D_8 configuration has length seven. These statements about G are kept separate from the full graph groups.

4 Full graph groups

For a prime $p \equiv \pm 1 \pmod 8$, choose $\epsilon \in \{1, -1\}$ with $p \equiv \epsilon \pmod 8$, and put $d = (p - \epsilon)/8$. The full group of a graph with parabolics S_4, S_4 and intersection D_8 is

$$\mathrm{Aut}\, X \cong \begin{cases} \mathrm{PSL}(2, p) \times C_2, & d \text{ even}, \\ \mathrm{PGL}(2, p), & d \text{ odd}. \end{cases}$$

An explicit torus-normalizer calculation constructs the required type interchange. Tutte's bound then gives index at most four over G. Simplicity forces G to be normal, and the centralizer and the action on the two S_4 classes determine which of the displayed groups occurs. Thus the printed PGL identifications at $p = 17, 31, 47, 79$ are incorrect; the identifications at $p = 23, 41, 71, 73, 89$ are recovered. In all these prime cases the full graph action still has arc length four.

For $q = 9$, the graph has the 15 duads and 15 synthemes on six letters as its two types. Its symplectic realization proves that it is the generalized quadrangle of order $(2, 2)$. A displayed permutation interchanges types and is checked directly on all 45 edges. Together with the faithful S_6 action this attains the Tutte bound of 1440. The full group is therefore $\mathrm{P\Gamma L}(2, 9)$, with full arc length five, as the source states. The construction and the named graph are classical; the literal duality certificate is a verification of the proof.

For $q = 17, 31, 79, 97$, suppressing the degree-two vertices in the S_4, D_{16} graph gives a simple cubic graph. The suppression proof excludes parallel edges and uses the actual transitive action. Tutte's bound gives index at most two, while the outer automorphism exchanges the two S_4 classes and cannot preserve the required action. The full group is $\mathrm{PSL}(2, q)$, recovering all four stated groups.

5 Preliminary formulas and literal corrections

The elementary order obstruction above proves L063 Lemma 4.4. For Lemma 4.5, an abelian transitive permutation group has the same stabilizer at every point. Faithfulness makes that stabilizer trivial, so the action is regular. If it is two-transitive, its trivial point stabilizer can be transitive on the other points only when there is one other point. Thus the degree is two. The word "permutation" here includes faithfulness; a group with a nonfaithful abelian action would first be replaced by its image.

The elementary abelian orbit formulas in Lemma 4.1 need their field parameter to be specified. This also explains why the source correctly declines to give a number of conjugacy classes from those formulas alone.

Lemma 1. *Let $0 \neq V \subset \mathbb{F}_q$ be an additive subgroup of order p^m, and let*

$$K = \{a \in \mathbb{F}_q : aV \subset V\} = \mathbb{F}_{p^k}.$$

Put $A_F = \mathbb{F}_q^*$ in characteristic two and $A_F = (\mathbb{F}_q^*)^2$ in odd characteristic. In $G = \mathrm{PSL}(2, q)$,

$$\mathrm{N}_G(U_V) = U_{\mathbb{F}_q} : (K^* \cap A_F).$$

If $1 \neq D \leq K^* \cap A_F$, then

$$\mathrm{N}_G(U_V : D) = U_V : (K^* \cap A_F).$$

Consequently the orbit lengths are respectively

$$\frac{q^2 - 1}{e(p^k - 1)}, \qquad \frac{(q^2 - 1)p^{n-m}}{e(p^k - 1)}, \quad q = p^n,$$

where $e = 2$ if p is odd and n/k is even, and $e = 1$ otherwise.

Proof. The set K is closed under addition and multiplication and contains one. Every nonzero element sends the finite set V bijectively to itself, so its inverse also lies in K. Thus K is a field, and k divides both n and m. A normalizer must fix the unique fixed point of U_V, hence has affine

form $x \mapsto ax + b$. It normalizes U_V exactly when $a \in K^* \cap A_F$; the translation b is unrestricted. For the second group U_V is characteristic as its unique Sylow subgroup for the characteristic. Conjugating any scalar $c \neq 1$ in D gives translation $(1-c)b$, which lies in V exactly when $b \in V$, since $(1-c)^{-1} \in K$. The displayed normalizers follow. Divide $|G| = q(q^2-1)/\gcd(2, q-1)$ by their orders and use the square criterion for K^* in $\mathbb{F}_q$ to obtain the formulas. □

Thus the source's k is the degree of this multiplier field, not an arbitrary common divisor. Different V can have different such fields; the formulas determine each orbit length, not the number of orbits.

Three additional preliminary corrections are explicit.

1. The odd-characteristic single-class clauses of Lemma 4.1(5) must require $d > 2$ when referring to D_{2d}. At $d = 2$ the four-groups belong to item (6). For example, at $q = 5$ the former formula would give 15 four-groups, whereas A_5 has five: each is the subgroup of double transpositions fixing one of the five letters. This verifies the correction without an ambient search.

2. The $\mathrm{PGL}(2, s)$ index in Table 8 is

$$[\mathrm{PSL}(2, s^2) : \mathrm{PGL}(2, s)] = \frac{q(q^2-1)}{2s(s^2-1)}, \qquad q = s^2.$$

   The printed denominator omits the factor two. This follows directly from the two group orders.

3. The printed exhaustive list in Proposition 4.6 omits the degree-two and degree-four actions of a maximal S_4 at prime q. For instance $S_4 < \mathrm{PSL}(2, 7)$ has its action on S_4/A_4 of degree two and its faithful natural action on S_4/S_3 of degree four. Its standalone clause lists degree three only; the separate PGL clauses impose $q = s^2$. The common action proof includes all three S_4 actions with their correct kernels. Assertions about representations inside one fixed subgroup are distinguished from the two ambient embedding classes.

L064's proof of Lemma 24 also incorrectly excludes every $\mathrm{PGL}(2, 3^w)$ from $\mathrm{PSL}(2, 3^{2m})$ for odd prime m. Both $w = 1$ and $w = m$ give subfield embeddings. The required two-transitive overgroup exclusion is nevertheless true and is proved by the common action theorem; it does not use that auxiliary assertion. In L063's proof of Lemma 5.3, Steps 4–5 confuse the labels A_4 and $E_{16} : 3$. The total number of the A_4 subgroups in the stated characteristic-two family is $q(q^2-1)/12$, rather than $q(q-1)/12$: the additional factor $q+1$ counts their unique fixed projective points. The accepted coordinate proof gives the correct intermediate counts while recovering the lemma's final formula. In three displayed transporter formulas of L064 Proposition 30, on physical pages 19, 20 and 21, the denominator must be the actual ambient order $|\mathrm{PSL}(2, p^{2n})|$, not $|\mathrm{PSL}(2, 2^{2n})|$. The literal normalizer proofs give the stated final values without that transcription error. In Section 6 of L063, the second path in the A_5, D_{30} argument starts at type one, and the S_4, D_{24} argument must say D_{24}, not D_{18}, when acting on cosets of D_8. The full local proofs in the collection fix both slips.

6 Separate attribution and source accounting

L063's Theorem 1.1 and its concluding completeness assertion are false as printed because of the omitted affine family. The corrected proof retains the valid configurations and all valid class

counts, then replaces the incomplete subfield argument and the unordered counts. The local arc columns are recovered, and the full graph groups have the stated four corrections. Preliminary borrowed results are identified as premises with the exact qualifications above; we do not claim to have rederived the classical subgroup classification.

L064 contains proofs for Lemma 12, Proposition 16, Lemmas 17–19, Proposition 20, Lemma 21, Proposition 22, Lemmas 23–25, Proposition 26, Lemmas 27–29 and Proposition 30. Its Lemma 21 lists the affine type without the published lemma's restriction, but Proposition 22 still restricts the configuration to extension degree two. The respective dispositions are therefore different. Its Propositions 16 and 26 share the unordered-count corrections, and Lemma 24 has the auxiliary correction just discussed. Its other surviving assertions follow from the same explicit arguments as L063.

The fixed appendix does not contain the advertised entire A_5 count proof, the standalone published $q = 25/S_4$ lemma, or Section 6's claimed arc proofs on appendix pages 19–20. Those pages actually concern Proposition 30. We supply the required proofs for L063 and do not invent missing L064 content or assign it extra claims. This is an edition boundary, not a statement that no other appendix edition ever existed.

The project contributions are the shared coordinate, transporter and normalizer arguments; explicit unordered orbit corrections; complete local stabilizer calculations; and the proofs and certificates identifying the full graph groups. Where an argument reuses a source proof, such as the fourth-arc obstruction in the S_4, D_{24} case, that attribution remains in its component. Distinct review records also disclose the collaborative contributions to the cubic group bounds and the Frobenius argument. Exact source ledgers and input bindings accompany the collection. No compression ratio, new open-conjecture answer or Lean verification is claimed in this completion. Those tasks remain outside the present intake stopping boundary.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* PSL$(2, q)$*: part II*, Ars Math. Contemp. **6** (2013), 365–388.

[2] The same authors, *Part II (Appendix)*, retained 21-page edition.

[3] J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* PSL$(2, q)$*: part I*, Ars Math. Contemp. **3** (2010), 177–192.

Subfield planes in projective rank two geometries

Human Compression project

Abstract

A plane over a subfield supplies a locally two-transitive coset geometry without requiring the ambient field to have degree two over that subfield. We give the construction and check its intersection, connectivity and primitivity directly. For $\mathrm{PSL}(2, 512)$ the two parabolics have orders 504 and 448, with intersection of order 56. This contradicts the exhaustion in the fixed Part II classification and the corresponding restriction in its separately catalogued appendix. It is one defect appearing in two documents, not two independent discoveries. No complete corrected classification is asserted.

1 Scalar subspaces and local actions

We write $V \rtimes K^*$ for the affine transformations $x \mapsto ax + v$, where K is a finite field, V is a vector space over K, $a \in K^*$ and $v \in V$. Its multiplication is $(v, a)(w, b) = (v + aw, ab)$. All actions on cosets below are allowed to have a kernel; this is also the convention in the source classification.

Lemma 1. *Let $W \leq V$ be K-subspaces, and set $B = W \rtimes K^*$. The subgroups between B and $V \rtimes K^*$ are precisely $T \rtimes K^*$ with $W \leq T \leq V$. Such a subgroup covers B if and only if $\dim_K(T/W) = 1$. In this case its action on the cosets of B induces $\mathrm{AGL}(1, K)$ and is two-transitive, with kernel W.*

Proof. If D contains B and $(v, a) \in D$, multiplying by $(0, a^{-1}) \in B$ shows that $(v, 1) \in D$. Thus $D = (D \cap V) \rtimes K^*$. The additive group $D \cap V$ is invariant under all nonzero scalars, so it is a K-subspace containing W. This proves the correspondence and the assertion about covers. Since W is normal in $T \rtimes K^*$, the induced coset action is the affine action of $(T/W) \rtimes K^*$ on T/W. If this quotient has dimension one, translations are regular and K^* is regular on its nonzero vectors. The action is therefore two-transitive and faithful on the quotient. Its kernel before passage to the quotient is exactly W. □

The distinction between an additive plane and a subfield is essential. For F/K of prime degree $m > 2$, there is no intermediate subfield of degree two. Nevertheless there are two-dimensional K-subspaces of F. Lemma 1 uses only a subspace.

2 The projective construction

Let $s = 2^n \geq 4$, let m be prime, and put $K = \mathbb{F}_s \subset F = \mathbb{F}_{s^m}$. In characteristic two every nonzero determinant has a square root, so $\mathrm{PGL}(2, F) = \mathrm{PSL}(2, F)$. We use fractional linear transformations as literal elements of this group. In particular the transformation $x \mapsto ax + b$ is represented by $\left(\begin{smallmatrix} a & b \\ 0 & 1 \end{smallmatrix}\right)$. Write $U_T = \{x \mapsto x + t : t \in T\}$ and $C = \{x \mapsto ax : a \in K^*\}$. The retained classical subfield theorem states that the natural $\mathrm{PSL}(2, K)$ is maximal in $\mathrm{PSL}(2, F)$ when m is prime. This is a subgroup theorem used by the source, not its geometry classification.

Theorem 2. *For every K-plane $T \subseteq F$ containing K, set*

$$G = \mathrm{PSL}(2, F), \qquad H = \mathrm{PSL}(2, K), \qquad Q = U_T C, \qquad B = U_K C.$$

Then $H \cap Q = B$, and the right-coset incidence system with parabolics H, Q is a thick, connected, flag-transitive rank two geometry. It is residually weakly primitive and locally two-transitive. Its two local degrees are $s + 1, s$, and

$$|H| = s(s^2 - 1), \qquad |Q| = s^2(s - 1), \qquad |B| = s(s - 1).$$

These assertions hold for every prime $m \geq 2$.

Proof. The set Q is a subgroup because T is stable under multiplication by K. Its elements have the unique affine expressions $x \mapsto ax+t$ with $a \in K^*$ and $t \in T$. Such an element belongs to H precisely when $t \in K$: a projective transformation over K fixing infinity is affine with both coefficients in K, and its normalized bottom right entry is one. Thus the intersection is exactly B. The displayed orders follow from these unique expressions and the classical order of $\mathrm{PSL}(2, K)$.

The action of H on H/B is its natural action on $\mathbb{P}^1(K)$. It is two-transitive: the stabilizer of infinity contains all translations of K, which are transitive on the other points. The action of Q on Q/B is two-transitive by Lemma 1, since $\dim_K(T/K) = 1$. The local degrees are consequently $s + 1$ and s. In particular both intersections are maximal in their parabolics. The second local action has the nontrivial kernel U_K; faithfulness of local actions is not a hypothesis of the source problem.

For completeness, take the elements of the incidence system to be the right cosets of H and Q, of distinct types, incident when they intersect. Each incident pair contains an element g and is therefore the translate of $\{H, Q\}$ by g. This proves flag transitivity. Its neighbours at the base vertices are the coset actions just computed, so every vertex lies on an edge and both residues are thick. Choose $t \in T \setminus K$. The translation by t lies in $Q \setminus H$. The retained maximality of H therefore gives $\langle H, Q\rangle = G$. Paths obtained by multiplying alternately by elements of H and Q show that the incidence graph is connected. In rank two this is the only connectedness requirement for residues of rank at least two.

Finally G acts primitively on G/H, since H is maximal, and both rank one residue actions are primitive, since they are two-transitive. These are exactly the weak primitivity and residual primitivity conditions in rank two. The whole action is faithful because its kernel is a normal subgroup of the simple group G contained in the proper subgroup H. □

There are $(s^{m-1} - 1)/(s - 1)$ choices of T for the fixed subfield and fixed Borel B: they correspond to the lines of the $(m - 1)$-dimensional space F/K. This counts subspaces, not isomorphism classes of geometries. For $m = 2$ there is only $T = F$, recovering the known quadratic family. For $m > 2$ every such plane is a proper additive subspace of F.

3 The two fixed sources

The first source is Buekenhout, De Saedeleer and Leemans, *On the rank two geometries of the groups* $\mathrm{PSL}(2, q)$*: part II*, Ars Mathematica Contemporanea 6 (2013), 365–388 [1]. Its Theorem 1.1 asserts exhaustion by Tables 1–6. On printed page 382, Lemma 5.11 restricts the affine overgroup $E_{2^{2n}} : (2^n - 1)$ to $m = 2$; Proposition 5.12 and Table 4 on page 367 use that restriction in the classification with subfield parabolic $\mathrm{PSL}(2, 2^n)$.

Corollary 3. *The exhaustion asserted by Theorem 1.1, Proposition 5.12 and Table 4, and the $m = 2$ restriction in Lemma 5.11 of this fixed edition, are false.*

Proof. Apply Theorem 2 with $s = 8$, $m = 3$. Then $q = 512$, $n = 3$, and the parabolics have orders $504, 448$ with intersection 56. The first row of Table 4 requires $m = 2$, and every other row requires $n = 1$ or $n = 2$. Tables 1–3 concern the maximal parabolics A_5, A_4, S_4, whose orders are different from both 504 and 448. Table 5 requires odd characteristic and Table 6 requires an odd square field. Thus none of the tables contains this geometry, even after interchanging its two types. Its local degrees $9, 8$ and the verified maximality and intersection satisfy the stated hypotheses. The same group Q is the forbidden affine overgroup in Lemma 5.11. □

The separately indexed appendix [2] is a fixed 21-page document with the placeholder heading *Ars Mathematica Contemporanea x (xxxx) 1–x*; its numbering differs from the published paper. On page 12, the proof headed "Proposition 22", general case, subcase 2, again admits this affine parabolic only for $m = 2$. The construction above contradicts that restriction and the resulting purported exhaustion. The proof of Lemma 21 on page 11 actually lists the affine abstract group without that restriction. We do not attribute a false restriction to that passage, nor infer the full statement of its missing lemma from the heading alone.

This is one omitted family shared by two catalogue entries. Other classification cases and numerical counts in those sources have not thereby been recovered or refuted. In particular no corrected global class count or full automorphism group is claimed. Both complete source texts were read to establish this boundary; only the indicated assertions are contradicted here.

4 Attribution and finite verification

The quadratic example and the analogous A_5 families are already present in the source. The general scalar-subspace correspondence is elementary; the present application uses the same mechanism as the project's earlier manuscript *Scalar subspaces in rank-two Suzuki geometries.* Here its role is to remove an unjustified extension-degree restriction. The natural projective action, the order formula, simplicity and subfield maximality are retained classical facts. This note neither claims historical priority nor asserts that no later correction exists.

The accompanying program independently represents $\mathbb{F}_{512}$ as a binary polynomial field, constructs the actual projective subgroups H, Q, B, and checks their orders, intersections and the two local actions. It also checks the quadratic boundary and the known $s = 4$ specialization. These finite controls supplement the proof; neither the uniform argument nor the retained subfield maximality theorem is inferred from a finite sample. No Lean verification or open-conjecture credit is asserted.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, On the rank two geometries of the groups PSL$(2, q)$: part II, *Ars Math. Contemp.* **6** (2013), 365–388. Fixed source.

[2] F. Buekenhout, J. De Saedeleer and D. Leemans, On the rank two geometries of the groups PSL$(2, q)$: part II (APPENDIX), the separately retained 21-page edition. Fixed appendix.

Two-transitive coset actions in dimension two

Human Compression Project

Abstract

We derive the two-transitive actions needed in the two fixed Part II originals from the classical subgroup classification of the linear groups of dimension two. A point stabilizer of order b has degree at most $b+1$. This bound, together with a short divisibility argument, removes the subfield alternatives and leaves a small list of exceptional actions. The resulting common argument also determines the possible overgroups of the small intersections used in those originals. We do not assume their Proposition 4.6 or its unpublished reference, and we do not count embeddings or pairs of parabolics here.

1 Premises and the elementary bound

We write D_{2n} for a dihedral group of order $2n$. The classical subgroup theorems of Dickson, Moore and Suzuki are borrowed premises: the proper subgroups of $\mathrm{PSL}(2,t)$ are cyclic or dihedral groups, elementary abelian groups and their scalar cyclic extensions, A_4, S_4, A_5, and the subfield groups $\mathrm{PSL}(2,s)$ and $\mathrm{PGL}(2,s)$. We use their usual field restrictions, maximal-subgroup lists and normalizers. In particular a subgroup $\mathrm{PGL}(2,s)$ in $\mathrm{PSL}(2,t)$, for odd s, requires an even extension degree. The normalizer of a four-group in odd characteristic contains an A_4. A torus has its dihedral normalizer. The two S_4 classes in $\mathrm{PSL}(2,7)$ and the two A_5 classes in $\mathrm{PSL}(2,9)$ and $\mathrm{PSL}(2,11)$ are interchanged by $\mathrm{PGL}(2,t)$. These are subgroup and fusion premises, not assertions about the geometries to be recovered. They are identified separately in Sections 4.1, 4.2 and 4.4 of the fixed published source [1]. We retain that source's attribution to [2, 3, 4]; we do not represent the historical books as newly checked source editions. The modern statement in [5, Theorem 2 and Lemmas 1–2], consulted separately, also records the subgroup types in odd characteristic and the small exceptional subgroup lattices. Its entire design theory is not an input to this proof.

We also use the classical isomorphisms $\mathrm{PSL}(2,4) \cong \mathrm{PSL}(2,5) \cong A_5$, $\mathrm{PSL}(2,7) \cong \mathrm{GL}(3,2)$, $\mathrm{PSL}(2,9) \cong A_6$ and $\mathrm{PGL}(2,5) \cong S_5$. The simplicity of $\mathrm{PSL}(2,t)$ for $t \geq 4$ is retained as a premise. The small subgroup lattices of A_4, S_4, A_5 also follow directly from their natural permutation actions; we use their maximal subgroup lists below.

Lemma 1. *Suppose that H acts two-transitively on the cosets of a proper subgroup B, and put $d = [H : B]$. Then B is maximal in H, $d-1$ divides $|B|$, and $|H| \leq |B|(|B|+1)$.*

Proof. A nontrivial system of blocks would prevent a point stabilizer from being transitive on the other points. Thus the action is primitive, and B is maximal. The B-orbit on the other $d-1$ points has length $d-1$, which divides $|B|$. Multiplying $d \leq |B|+1$ by $|B|$ proves the bound. Faithfulness is unnecessary. □

2 The simple groups

Theorem 2. *Let $H = \mathrm{PSL}(2,t)$, where $t \geq 4$ is a prime power. The two-transitive coset actions are the natural action with point stabilizer $E_t : C_{(t-1)/(2,t-1)}$, and the following exceptional actions, with repetitions under the small isomorphisms suppressed:*

H	B	$[H : B]$
A_5	A_4	5
A_5	D_{10}	6
$\mathrm{PSL}(2,7)$	S_4	7
$\mathrm{PSL}(2,9)$	A_5	6
$\mathrm{PSL}(2,11)$	A_5	11.

Here each embedding class of the displayed stabilizer gives its corresponding coset action. The assertion is not a count of ambient embeddings into a larger group.

Proof. The kernel is normal in the simple group H and is contained in the proper subgroup B, hence is trivial. Apply the maximal-subgroup list and Lemma 1.

For a torus normalizer the two possible degrees are $t(t-1)/2$ and $t(t+1)/2$. For odd $t \geq 7$, their point stabilizers have order at most $t+1$, whereas $t(t-1)/2 - 1 > t+1$. For even $t \geq 8$, their order is at most $2(t+1)$ and $t(t-1)/2 - 1 > 2(t+1)$. Thus only $t = 4, 5$ remain. In A_5, D_{10} has index six; D_6 has index ten, which fails $9 \mid 6$.

If $B = A_4$, the order bound is $|H| \leq 156$, so $t \leq 6$ in odd characteristic and $t \leq 5$ in even characteristic. Only $t = 4, 5$ give the required subgroup. If $B = S_4$, the bound is $|H| \leq 600$. The only possible fields with an S_4 subgroup are $t = 7, 9$; for $t = 9$ the degree is fifteen and $14 \nmid 24$. If $B = A_5$, the bound is $|H| \leq 3660$. Apart from $H = A_5$, the possible orders divisible by sixty are those for $t = 9, 11, 19$; the last has degree fifty-seven and $56 \nmid 60$. These finite checks use the order $t(t^2-1)/(2, t-1)$ and the prime powers in the stated bounds.

For an even field, a maximal proper subfield group has $t = s^r$ with r prime. At $r = 2$ its index is $s(s^2+1)$ and its order is $s(s^2-1)$, so the index minus one already exceeds its order. The index increases with r. For an odd field and a maximal simple subfield group, $r \geq 3$. Its index is at least $s^2(s^4 + s^2 + 1)$, again exceeding its order plus one. The remaining maximal subfield possibility is $\mathrm{PGL}(2,s)$ in $\mathrm{PSL}(2,s^2)$ for odd $s \geq 3$. Its order is $b = s(s^2-1)$ and its index is $d = s(s^2+1)/2$. But

$$1 < \frac{b}{d-1} = 2 - \frac{4s-4}{s^3+s-2} < 2.$$

Consequently $d - 1$ cannot divide b. This removes every proper subfield alternative.

The listed actions do exist. Translations in a Borel subgroup are transitive on the finite points of the projective line, proving the natural case. The natural A_5 action on five letters is two-transitive. Its action on the six Sylow five-subgroups is two-transitive because a Sylow five-subgroup fixes only itself: another fixed point would put it in a different normalizer of order ten. It therefore has one orbit on the other five points. The action of $\mathrm{GL}(3,2)$ on its seven nonzero vectors is two-transitive, since two distinct such vectors are independent. The natural A_6 action gives the degree-six case.

For $H = \mathrm{PSL}(2,11)$ choose one class of A_5 subgroups. It has eleven members. There are fifty-five A_4 subgroups, all conjugate and self-normalizing; each chosen A_5 contains five. Double counting shows that each A_4 lies in exactly one member of the chosen A_5 class. Hence distinct members do not intersect in an A_4. In the action on that class the stabilizer A_5 has no additional

fixed point, because the normalizer of an A_5 is itself. Its possible nontrivial orbit lengths at most ten are five, six and ten, by the subgroup orders of A_5. Length five would give an A_4 intersection and is impossible. Ten remaining points cannot be partitioned into orbits of length six. They form one orbit of length ten. This proves two-transitivity, for either of the two A_5 classes. □

3 The projective groups

Theorem 3. *Let $H = \mathrm{PGL}(2,t)$ for odd $t \geq 5$. Its two-transitive coset actions have point stabilizer $\mathrm{PSL}(2,t)$ and degree two, the natural Borel $E_t : C_{t-1}$ and degree $t+1$, or, when $t = 5$, S_4 and degree five. For $\mathrm{PGL}(2,3) = S_4$ the stabilizers are A_4, D_8, S_3, of degrees two, three and four.*

Proof. Put $N = \mathrm{PSL}(2,t)$. If N is in the kernel, the nontrivial action factors through $H/N = C_2$, giving the first case. Otherwise a nontrivial kernel would meet the simple group N trivially and centralize it. The centralizer of N in H is trivial: commuting with all translations fixes their unique common fixed point and gives an affine map with scalar one; commuting also with the opposite unipotent subgroup makes its translation zero. Thus the action is faithful.

Primitivity now makes N transitive. For a point stabilizer B write $C = B \cap N$. Then $H = NB$, $|B| = 2|C|$, and $d = [H : B] = [N : C]$. In particular

$$d - 1 \mid 2|C|, \qquad |N| \leq |C|(2|C| + 1). \tag{1}$$

We apply the full subgroup list to C, which need not be maximal in N.

If C has nontrivial unipotent radical $O_p(C)$, this radical is characteristic in C and has a unique common projective fixed point. Since $C \lhd B$, B fixes this point. Maximality of B in H makes B the full Borel, giving the natural action. This covers all affine types, including nontrivial unipotent cyclic subgroups.

If C is a nontrivial semisimple cyclic group, B normalizes its torus. For an involution this follows by centralizing it, which preserves its two eigenlines as an unordered pair. If C is dihedral with rotation group of order greater than two, that rotation group is characteristic, with the same conclusion. Thus maximality forces B to be a torus normalizer. Its order is at most $2(t+1)$ and the smaller possible degree is $t(t-1)/2$. For $t \geq 7$, $d - 1 > 2(t+1)$. At $t = 5$ the two degrees are ten and fifteen, with stabilizer orders twelve and eight, and both fail divisibility. The case $C = 1$ fails (1). If C is a four-group, its normalizer contains an A_4, while $|B| = 8$. This normalizer is proper, since C cannot be normal in the simple group N. It would strictly contain the maximal subgroup B, a contradiction.

If $C = A_4$, (1) gives $|N| \leq 300$. Only $t = 5, 7$ can occur; $t = 7$ gives $d = 14$ and $13 \nmid 24$. For $t = 5$ the extension is S_4 inside S_5, with its natural degree-five action. If $C = S_4$, the bound is $|N| \leq 1176$. The possible fields are $t = 7, 9$; $t = 9$ fails $14 \mid 48$. At $t = 7$, an element of $B \setminus N$ would normalize the S_4 subgroup, contrary to the interchange of its two N-classes by H. If $C = A_5$, the bound is $|N| \leq 7260$. The proper possibilities are $t = 9, 11, 19$; $t = 19$ fails $56 \mid 120$. For $t = 9, 11$, interchange of the two A_5 classes again excludes such a B. The case $C = N = A_5$ belongs to the already separated degree-two action.

Finally suppose C is a proper subfield group with field size s. For $C = \mathrm{PSL}(2,s)$ the smallest possible index occurs at $t = s^2$ and equals $s(s^2+1)$; this already exceeds $2|C|+1 = s(s^2-1)+1$. For $C = \mathrm{PGL}(2,s)$, t is an even power of s. At $t = s^2$,

$$3 < \frac{2|C|}{d-1} = 4 - \frac{8s-8}{s^3+s-2} < 4 \qquad (s \geq 3),$$

so divisibility fails. At any larger allowed extension degree, even the index for $t = s^3$ is $s^2(s^4 + s^2 + 1)/2 > 2s(s^2 - 1) + 1$, and the bound fails. This exhausts C.

The Borel and degree-two actions plainly exist, and the exceptional degree-five action is the natural S_5 action. The maximal subgroups A_4, D_8, S_3 of S_4 give respectively its quotient action on two points, its S_3 quotient on three points, and its natural action on four points. They exhaust the case $t = 3$. □

4 Solvable overgroups

The following elementary observations handle the remaining Dickson types without a classification of general two-transitive groups.

Lemma 4. *For a dihedral group, a two-transitive coset action has degree two or three. In degree two the stabilizer is any subgroup of index two; in degree three it is a dihedral subgroup of index three.*

Let $H = V : C$ be a scalar affine subgroup in characteristic p, where $V \neq 0$ is elementary abelian and C is cyclic of order prime to p. A two-transitive coset action either has degree two and its stabilizer contains V, or has a stabilizer conjugate to $W : C$, with W a C-invariant subgroup of V and

$$[V : W] = |C| + 1.$$

The induced scalar action on $V/W \setminus \{0\}$ is regular.

Proof. For a dihedral group, the orbits of its normal rotation group form blocks. If this group is in the kernel, the action factors through C_2. Otherwise it is transitive in the faithful quotient. A transitive abelian permutation group is regular, since all point stabilizers coincide and hence lie in the kernel. The point stabilizer in the resulting dihedral permutation group therefore has order at most two. Transitivity on the other points implies that the degree is at most three. A subgroup of index three with a transitive rotation group contains a reflection and is dihedral. Conversely the quotients on two and three points are C_2 and S_3, respectively.

For the affine group, primitivity likewise says that V is in the kernel or is transitive in the faithful quotient. In the first case the action factors through the abelian group C; two-transitivity then forces degree two. In the second case put $W = V \cap B$. Regularity of the image of V shows that W is the kernel within V, hence is H-invariant, and $H = VB$. The image of B in $H/V = C$ is all of C. Modulo W, the subgroup B/W is a complement to V/W. Such complements are conjugate by V/W: one may average an affine fixed point over the cyclic complement, since $|C|$ is invertible in characteristic p. Thus B is conjugate to $W : C$.

For a nonidentity scalar c, multiplication by $c - 1$ is invertible on V and maps the invariant finite subspace W onto itself. It is therefore invertible on V/W. Thus the nonidentity elements of C have no nonzero fixed vector in V/W, and two-transitivity says that this free action on the nonzero vectors is also transitive. This proves the formula. □

5 The small intersections in the fixed originals

Corollary 5. *Within the field scopes used in the published Part II source, the possible proper two-transitive overgroups of the indicated intersection are as follows. Each listed dihedral overgroup*

requires its usual torus divisibility condition.

B	*ambient field scope*	H
D_{10}	*the maximal* A_5 *cases*	D_{20}, D_{30}, A_5
A_4	*the maximal* A_5 *cases*	$S_4, A_5, E_{16} : C_3$
D_6	q *prime,* $q \equiv \pm 1 \pmod 8$	D_{12}, D_{18}, S_4
D_8	q *prime,* $q \equiv \pm 1 \pmod 8$	D_{16}, D_{24}, S_4
A_4	q *prime,* $q \equiv \pm 1 \pmod 8$	S_4, A_5
A_5	$q = 9^m$ *or* 11^m, m *odd prime*	$\mathrm{PSL}(2,9)$ *or* $\mathrm{PSL}(2,11)$
S_4	$q = 7^m$, m *odd prime*	$\mathrm{PSL}(2,7)$
D_8	$q = 9$	S_4
S_4	$q = 25$	$\mathrm{PGL}(2,5)$.

The last two alternatives in the A_5 row are restricted to their actual defining characteristic. The affine $E_{16} : C_3$ alternative occurs only in characteristic two and needs at least two dimensions over $\mathbb{F}_4$ in the ambient additive field. An S_4 or A_5 alternative is included only when it embeds in the specified ambient group.

Proof. For the cyclic and dihedral Dickson types, a nonabelian stabilizer cannot lie in a cyclic group. Lemma 4 gives exactly the displayed double and triple dihedral orders. For $q = 9$ neither D_{16} nor D_{24} embeds: the maximal tori have orders four and five.

Consider an affine overgroup. Its stabilizer either contains the whole additive kernel with index two, or is $W : C$ as in Lemma 4. For $B = A_4$, the latter forces $W = E_4$, $C = C_3$, and $[V : W] = 4$, giving $E_{16} : C_3$. An index-two extension would instead require a faithful scalar C_6 on E_4, impossible since its scalar group has order three. This also shows directly that no affine alternative exists in odd characteristic for $B = A_4$.

For $B = D_{10}$, a nontrivial additive normal subgroup must be C_5. The transitive-kernel alternative would demand an additive quotient of size $|C_2|+1 = 3$ in characteristic five, impossible. The index-two alternative would be $C_5 : C_4$. In the present maximal-A_5 scope a characteristic-five field has size 5^m with m odd, so its scalar torus has order $(5^m - 1)/2$, which is twice an odd number. It contains no C_4. For $B = D_6$ the affine possibility would require characteristic three; the stipulated prime $q \equiv \pm 1 \pmod 8$ is not three. For D_8 in odd characteristic there is no nontrivial normal subgroup of odd prime-power order. Finally S_4 and A_5 cannot occur as these affine stabilizers: in odd characteristic they have no nontrivial normal subgroup of defining prime-power order, and they are not cyclic. Thus no affine cases were lost.

The small exceptional overgroups are read directly from the maximal-subgroup lists: A_4 has its degree-four action with stabilizer C_3; S_4 has the three actions in Theorem 3; A_5 has the degree-five and degree-six actions in Theorem 2. These give precisely the S_4 and A_5 entries above.

For all larger subfield overgroups, apply Theorems 2 and 3 to the abstract overgroup. Reading off the displayed stabilizers gives $\mathrm{PSL}(2,7)$ above S_4, $\mathrm{PSL}(2,9)$ and $\mathrm{PSL}(2,11)$ above A_5, and $\mathrm{PGL}(2,5)$ above S_4 or its normal A_5. Such groups are neither cyclic, dihedral, affine nor one of the three small exceptional Dickson types, so their embeddings must be subfield embeddings in the actual defining characteristic. This excludes $\mathrm{PGL}(2,5)$ from the characteristic-three and -eleven A_5 rows, and from the characteristic-seven S_4 row. It excludes $\mathrm{PSL}(2,7)$ in characteristic five. The natural Borel cases give no further small stabilizers. This proves the table. □

6 Source use and limits

Corollary 5 proves the overgroup possibilities used in Lemmas 5.1, 5.2, 5.7–5.9, 5.14, 5.15, 5.17 and 5.20 of the fixed published original L063. It supplies the common argument for Lemmas 12, 17–19, 24, 25 and 27 of the retained appendix L064, with the published parameter restrictions stated explicitly. The retained appendix begins at Lemma 12; it does not contain the advertised proof of published Lemma 5.1.

Theorem 2 and Theorem 3 replace the simple and projective portions of the unpublished classification cited as Proposition 4.6. They may be used in the separate generic subfield argument. The cyclic, dihedral and affine arguments explain their solvable counterparts, but no claim is made here that every literal assertion about uniqueness or class fusion in that proposition is correct. In particular a representation of an abstract subgroup and an embedding of that subgroup in a larger group are different counting problems.

The small bounds are finite integer calculations, not a group census. The classical subgroup classifications, their field restrictions, the stated exceptional isomorphisms and fusion facts remain attributed prior results. This manuscript supplies the common elimination and the elementary two-transitivity arguments. It does not establish all configuration counts, complete either original, solve an open conjecture, or claim Lean verification or a compression factor.

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Subfield planes and unordered pairs of parabolics

Human Compression Project

Abstract

We count two families from the fixed Part II classification of rank two geometries for $\mathrm{PSL}(2,q)$. In characteristic two, planes through a fixed subfield line give the parameter space; Frobenius gives the equivalence relation. This recovers the numerical formulas for the $A_5, E_{16} : 3$ family and extends its parameter description to the previously omitted subfield covers. In characteristic three, unordered pairs in an additive quotient give the family with two A_4 parabolics. This halves its printed conjugacy count under the source's stated convention, while preserving its automorphism count. Classical subgroup and automorphism theorems remain explicit premises.

1 Local actions and the counting convention

Let G be a finite simple projective group. For distinct proper subgroups H, Q, put $B = H \cap Q$. The rank two coset geometry has the cosets of H, Q as its two types, with nonempty intersection defining incidence. The two local actions are those of H, Q on their cosets of B. When both are two-transitive and one parabolic is maximal in G, these subgroups generate G and give a connected, flag-transitive, residually weakly primitive geometry. We use the proof of this assertion in the previously accepted projective-line manuscript; it depends only on coset actions and maximality.

The fixed Part II source, Section 2 on physical page 4, defines conjugacy by equality of the *sets* $\{H^g, Q^g\}$ and $\{H', Q'\}$. Thus interchanging the two types is allowed. We call this the unordered convention. We will give ordered counts separately when they differ. Equivalence under $\mathrm{Aut}(G)$ uses the same convention.

We retain the following classical facts used by the source: the natural subfield embeddings and their maximality at prime extension degree; the conjugacy of those embeddings in the cases specified below; $\mathrm{Aut}(\mathrm{PSL}(2,p^f)) = \mathrm{P\Gamma L}(2,p^f)$ for the ambient groups used here; and the usual projective orders. These are not assumed consequences of the geometry classification under audit. Every normalizer needed for the new counts is proved below.

2 Planes through a subfield line

Let $K = \mathbb{F}_s \subset F = \mathbb{F}_{s^m}$, with $s \geq 4$ even and m prime. Write $U_T = \{x \mapsto x + t : t \in T\}$ and $C = \{x \mapsto ax : a \in K^*\}$. Set

$$G = \mathrm{PSL}(2,F) = \mathrm{PGL}(2,F), \quad H = \mathrm{PSL}(2,K), \quad B = U_K C.$$

For a two-dimensional K-subspace T containing K, put $Q_T = U_T C$. The earlier accepted subfield-plane proof establishes

$$H \cap Q_T = B, \qquad |H| = s(s^2-1), \quad |Q_T| = s^2(s-1), \quad |B| = s(s-1),$$

and proves all geometry hypotheses. Its argument is a scalar-subspace correspondence: groups between $U_K C$ and $U_T C$ are the groups $U_W C$ for K-subspaces $K \subseteq W \subseteq T$. A cover has quotient dimension one and induces $\mathrm{AGL}(1, K)$ on its local residue. No subfield of degree two is needed.

Lemma 1. *The normalizer of B in G is B. For each conjugate of B, there is exactly one conjugate of H containing it. The conjugacy classes of the displayed geometries are consequently indexed by $\mathrm{PG}(m-2, s)$, the lines of F/K.*

Proof. The translation subgroup U_K is the unique Sylow two-subgroup of B, and its common projective fixed point is infinity. A normalizer of B is therefore affine, say $x \mapsto ax+b$. Normalizing U_K requires $aK = K$, hence $a \in K^*$. Conjugating any nonidentity scalar in C then requires $(1-c)b \in K$, so $b \in K$. This proves the normalizer assertion, including its converse.

The subfield groups H form one G-class in this even-characteristic case. Their point stabilizers form one class inside H. If H^g contains B, conjugate a point stabilizer inside H before applying g; the resulting transporter normalizes B. It belongs to B by the first paragraph, so $H^g = H$.

To compare two displayed pairs, the unequal parabolic orders first force H to be preserved. Their intersections then force B to be preserved. The transporter is in $N_G(B) = B$. Its translations lie in $K \subset T$ and its scalars in K^*, so it normalizes each Q_T. Hence two planes give conjugate pairs exactly when they are equal. The planes containing K correspond to the lines of F/K, as asserted. □

Proposition 2. *There are exactly $(s^{m-1}-1)/(s-1)$ conjugacy classes in this family. Their $\mathrm{Aut}(G)$-classes are the orbits of the absolute Frobenius $x \mapsto x^2$ on $\mathrm{PG}(m-2, s)$.*

Proof. The conjugacy count is Lemma 1. Every field automorphism preserves the standard H and B. Write an arbitrary ambient automorphism as an inner automorphism followed by a field automorphism. If it transports one of our pairs to another, its inner part must normalize B and therefore belongs to B. This part acts trivially on the planes. Conversely every Frobenius power gives the stated transport. This proves the orbit description, with no assertion that the field automorphism action is free. □

Lemma 3. *An affine subgroup Q containing B with B maximal in Q is one of the groups Q_T above. In particular every such affine extension has a two-transitive local action.*

Proof. Write $V = Q \cap U_F$ for its translations and D for its image in F^*. Both V and K are stable under K^*, so V is a K-space. The subgroup $U_V C$ lies between B and Q. If $V > K$, maximality forces $Q = U_V C$, and the scalar correspondence forces $\dim_K(V/K) = 1$. If $V = K$, an element $x \mapsto ax + b$ of Q normalizes U_K, so $a \in K^*$. Multiplying by its scalar inverse in C leaves translation by $b \in V = K$. Thus $Q = B$, contrary to the proper extension. This excludes the second case. □

3 The $A_5, E_{16} : 3$ formulas

Specialize to $K = \mathbb{F}_4$, $F = \mathbb{F}_{4^r}$ with r prime. The numbers in this section are the ones asserted by Lemmas 5.3–5.4 and the corresponding rows of Table 1 in the fixed source.

Lemma 4. *For odd prime r, the groups isomorphic to $E_{16} : 3$ in G form $(4^{r-1}-1)/15$ conjugacy classes.*

Proof. Such a group has characteristic normal subgroup E_{16}, whose nonidentity elements are unipotent and have a unique common fixed projective point. Put this point at infinity. Its complement of order three has a second fixed point; a translation puts that point at zero. The group is now $U_T C$ for a K-plane $T \subset F$. Conversely each such plane supplies the stated group. Affine conjugation identifies exactly the scalar multiples of the planes: translations can change a chosen complement but cannot change the translation subgroup.

The multiplier ring $E_T = \{a \in F : aT \subseteq T\}$ is a field. Closure under addition and multiplication is immediate; for nonzero $a \in E_T$, multiplication is a bijection of the finite set T, so $a^{-1} \in E_T$. It contains K, and $[E_T : K]$ divides both $\dim_K T = 2$ and $[F : K] = r$. Thus $E_T = K$. Every scalar orbit has length $(4^r - 1)/3$. There are

$$\begin{bmatrix} r \\ 2 \end{bmatrix}_4 = \frac{(4^r - 1)(4^r - 4)}{(16 - 1)(16 - 4)}$$

planes, by counting ordered independent pairs and dividing by the number of ordered bases of a plane. Division by the orbit length gives $(4^{r-1} - 1)/15$. □

Theorem 5. *For $r = 2$, the family with parabolics $A_5, E_{16} : 3$ has one class under each equivalence. For $r = 3$ it has five conjugacy classes and two $\mathrm{Aut}(G)$-classes. For prime $r > 3$ the respective counts are*

$$N = \frac{4^{r-1} - 1}{3}, \qquad \frac{N + (2^{r-1} - 1)}{2r} = \frac{2(4^{r-2} - 1) + 3 \cdot 2^{r-2}}{3r}.$$

Proof. The planes through K form $\mathrm{PG}(r - 2, 4)$, of size N; when $r = 2$ there is just the plane F. Put $V = F/K$, and let σ be squaring on this quotient. We must count its orbits, by Proposition 2. It has order dividing $2r$. Put $\tau = \sigma^2$, the K-linear Frobenius of order r, and $\beta = \sigma^r$, an involution semilinear over $K/\mathbb{F}_2$.

First suppose $r > 3$. If a line is fixed by τ, choose a nonzero vector v on it with $\tau v = av$, $a \in K^*$. Iterating gives $a^r = 1$, so $a = 1$ because r is prime and different from three. Writing $v = x + K$ gives $x^4 - x = c \in K$. The relative trace from F to K annihilates the left side and takes c to $rc = c$, so $c = 0$. Then $x \in K$, a contradiction. Every nonidentity power of τ has the same fixed lines. An odd power of σ other than β generates the cyclic group of order $2r$, so it has no fixed line either.

It remains to count the lines fixed by β. If $\beta(v) = av$, then $a \in K^*$ and one may replace v by $a^{-1}v$ to make it fixed: $\beta(a^{-1})a = a^{-1}$ since $a^3 = 1$. Each stable K-line therefore contains exactly one nonzero fixed vector. The fixed quotient space V^β is naturally $\mathbb{F}_{2^r}/\mathbb{F}_2$. Indeed, for a fixed class $x + K$, $c = \beta x - x \in K$ satisfies $\beta c = c$, hence $c \in \mathbb{F}_2$. The map $k \mapsto k^2 - k$ from K onto $\mathbb{F}_2$ provides k with $\beta k - k = c$; subtracting k gives a representative fixed by β. Its ambiguity is exactly $\mathbb{F}_2$. Thus β fixes $2^{r-1} - 1$ projective points. Burnside's formula gives the claimed average over $2r$ powers.

For $r = 3$, V has dimension two over K. The trace argument still shows that τ has no fixed nonzero vector. Since $\tau^3 = 1$, it is semisimple and its possible eigenvalues are the two nonidentity cube roots in K. The commuting semilinear map β squares eigenvalues and is invertible, so both eigenspaces occur and have the same dimension. Each therefore has dimension one. These are precisely the two invariant lines, and β interchanges them. The odd generators σ, σ^5 also interchange them and fix no line. The preceding fixed-quotient argument gives three lines fixed by β. The six fixed-point counts are $5, 0, 2, 3, 2, 0$, whose average is two. □

4 Two A_4 parabolics in characteristic three

Let $q = 3^m$, where m is an odd prime, and put $G = \mathrm{PSL}(2, q)$, $H = \mathrm{PSL}(2, 3) = A_4$, $B = U_{\mathbb{F}_3} \cong C_3$. The classical subfield theorem makes H maximal in G; its conjugates form one class. Put $V = \mathbb{F}_q/\mathbb{F}_3$, regarded as an additive group.

Lemma 6. *The conjugates of H containing B form a regular V-set under translations. Distinct such subgroups intersect exactly in B. Conjugacy classes of ordered pairs of them are indexed by $V \setminus 0$; unordered classes are indexed by $(V \setminus 0)/\{v \sim -v\}$.*

Proof. Any normalizer of B fixes its unique projective fixed point and is affine. Its slope must lie in $\mathbb{F}_3^* = \{1, -1\}$. An affine transformation belongs to $\mathrm{PSL}(2, q)$ precisely when its slope is a square in $\mathbb{F}_q$. Since m is odd, $q \equiv 3 \pmod 4$ and -1 is not a square. Therefore $N_G(B) = U_{\mathbb{F}_q}$.

Every copy of B inside H is conjugate under H, and $N_H(B) = B$. Thus the copies of H containing the fixed B are exactly its translates $H_t = u_t H u_t^{-1}$, with equality precisely when $t - t' \in \mathbb{F}_3$. Here we also use $N_G(H) = H$, which follows from maximality and simplicity. Translation acts regularly on the index set V. Since B is maximal in A_4, any two distinct H_t meet in B.

Every eligible pair of A_4 subgroups with intersection of order three can be transported to this set, since its intersection is a conjugate of B. Any transporter between two such pairs must normalize their intersection. Hence it acts as translation on V. Translating an ordered pair (a, b) to $(0, b-a)$ gives precisely its nonzero difference. Interchanging the entries changes this difference to its negative. These are all identifications, proving both parameterizations. □

Theorem 7. *Under the unordered convention of the fixed source, the number of conjugacy classes of these geometries is*

$$\frac{3^{m-1} - 1}{2}.$$

The number of $\mathrm{Aut}(G)$*-classes is two if $m = 3$, and $(3^{m-1} - 1)/(2m)$ otherwise. The ordered conjugacy count is $3^{m-1} - 1$.*

Proof. Distinct pairs give the required geometries: their intersection is C_3, each local A_4 action on four cosets is two-transitive, and maximality of either A_4 gives connectedness and the residual primitivity conditions. In characteristic three a nonzero vector is different from its negative. Lemma 6 therefore proves the two conjugacy counts.

In $\mathrm{P\Gamma L}(2, q)$ the normalizer of B consists exactly of affine maps with slope ± 1, followed by field automorphisms. The standard H is preserved by sign and by every field automorphism. It follows that the additional action on differences is generated by $v \mapsto -v$ and $\phi : v \mapsto v^3$. Its group is $C_2 \times C_m$. Inversion from interchanging the two parabolics is already in this group.

Suppose first that $m \neq 3$. If $\phi(x + \mathbb{F}_3) = x + \mathbb{F}_3$, then $x^3 - x = c \in \mathbb{F}_3$. Relative trace gives $mc = 0$, whence $c = 0$ and $x \in \mathbb{F}_3$. Thus no nonzero vector is fixed by ϕ, or by any of its nonidentity powers. A map $-\phi^k$ fixes no nonzero vector either: its mth power is negation. Consequently the action on $V \setminus 0$ is free, proving the formula.

For $m = 3$, V has dimension two over $\mathbb{F}_3$. The induced ϕ has order dividing three and is not the identity. To see the latter, if every $x \in \mathbb{F}_{27}$ satisfied $x^3 - x \in \mathbb{F}_3$, the additive map $x \mapsto x^3 - x$ would have image of size at most three, whereas its kernel has size three and its image has size nine. Thus ϕ has one nontrivial unipotent Jordan block on V, and its fixed subspace has dimension one. The two nonidentity powers of ϕ each fix two nonzero vectors. The three maps $-\phi^k$ fix none. Burnside's formula gives $(8 + 2 + 2)/6 = 2$. □

5 Source joins and attribution

Theorem 5 recovers the $A_5, E_{16} : 3$ counts printed in Table 1, rows 2–4, and Lemmas 5.3–5.4 of the published Part II paper. Lemma 4 separately counts subgroup classes, not geometries; these two counts must not be confused. The same mechanism supplies the parameter space for the omitted higher-extension subfield covers already proved in the accepted correction. It does not repeat that acquisition.

Theorem 7 corrects the inner conjugacy column in Table 5: the printed $3^{m-1} - 1$ becomes $(3^{m-1} - 1)/2$, and the printed eight at $q = 27$ becomes four. Both printed automorphism counts remain correct. The separately fixed 21-page appendix repeats the ordered count in its proof of Proposition 26, physical page 15. Its calculation counts the possible second parabolics after choosing a first one; it omits the additional identification required by the source's unordered definition. The explicit conjugator u_{-t} carries the unordered pair $\{H_0, H_t\}$ to $\{H_{-t}, H_0\}$. This gives the missing identification inside G, without a computational assumption.

The original numerical formulas recovered here belong to Buekenhout, De Saedeleer and Leemans. Gaussian counts, finite-field trace, Burnside's formula and normalizer actions are standard. The contribution of this note is the common projective parameter description, its extension to the already established missing covers, and the explicit audit of the unordered convention. No historical priority beyond this fixed-source comparison, complete treatment of either original, Lean verification, new open-conjecture answer or compression measurement is claimed.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* PSL$(2, q)$*: part II*, Ars Mathematica Contemporanea **6** (2013), 365–388.

[2] The same authors, *Part II (APPENDIX)*, the fixed separately catalogued 21-page edition, proof of Proposition 26.

[3] Human Compression Project, *Subfield planes in projective rank two geometries*; and *Local actions in rank two over a projective line*, previously accepted components.

Counting configurations above a cyclic subgroup of order three

Human Compression Project

Abstract

One regular action of a dihedral quotient counts all configurations whose maximal parabolic is A_4 and whose intersection has order three. Its reflections describe the second parabolics S_3, while its nonidentity elements describe ordered pairs of A_4 parabolics. Passing to unordered pairs identifies an element with its inverse. This recovers five rows of the source table and corrects the conjugacy counts in its last two rows. The full automorphism counts in those rows remain valid.

1 The hypotheses and the common normalizer

Let $p > 3$ be prime such that A_4 is maximal in $G = \mathrm{PSL}(2, p)$. The classical maximal-subgroup list gives exactly $p = 5$ or $p \equiv 3, 13, 27, 37 \pmod{40}$. In particular $p \equiv \pm 3 \pmod 8$. Put $E = \mathrm{PGL}(2, p) = \mathrm{Aut}(G)$, and choose $\epsilon \in \{1, -1\}$ with $3 \mid p - \epsilon$. Write

$$n = \frac{p - \epsilon}{2} = 3m, \qquad m = \frac{p - \epsilon}{6}.$$

Fix $B \cong C_3$ in G. We retain the classical facts that all such B are conjugate, that the subgroups A_4 form one G-class, and that $\mathrm{N}_E(A_4) \cong S_4$. The latter group acts on its normal A_4 by the usual automorphisms. Also, the centralizer torus of B has order n in G and $2n$ in E, and its normalizers are dihedral. These are subgroup facts, not assumptions about the number of geometries.

Choose torus and reflection generators so that

$$\mathrm{N}_E(B) = \langle R, S \mid R^{2n} = S^2 = 1,\ SRS = R^{-1} \rangle, \quad \mathrm{N}_G(B) = \langle R^2, S \rangle, \quad B = \langle R^{2m} \rangle.$$

After quotienting by B, write

$$\overline{N} = \mathrm{N}_E(B)/B = \langle x, s \mid x^{2m} = s^2 = 1,\ sxs = x^{-1} \rangle, \qquad K = \mathrm{N}_G(B)/B = \langle r = x^2, s \rangle.$$

Thus $|K| = 2m$ and r has order m. This presentation also covers $m = 1, 2$ without an exceptional convention for a small dihedral group.

Lemma 1. *The set $\mathcal{A}$ of subgroups A_4 containing B is a regular K-set. Choose one of them, denoted M, as the identity point. The stabilizer of M in $\overline{N}$ is generated by an external reflection j. Coordinates may be chosen with $j = xs$; its action on the K-coordinates is the automorphism*

$$\theta(r^k) = r^{-k}, \qquad \theta(r^k s) = r^{1-k} s.$$

Proof. All subgroups C_3 in A_4 are conjugate internally, and their normalizer in A_4 is C_3. Any transporter between two members of $\mathcal{A}$ can therefore be adjusted inside the target A_4 to normalize B. Hence $\mathrm{N}_G(B)$ is transitive on $\mathcal{A}$. Since M is maximal in the simple group G, it is self-normalizing. The stabilizer of M is consequently $\mathrm{N}_G(B) \cap M = \mathrm{N}_M(B) = B$. Moreover B

normalizes every member of $\mathcal{A}$, since it is contained in each. The action therefore factors through K and is regular.

The stabilizer in $\mathrm{N}_E(B)$ is $\mathrm{N}_{\mathrm{N}_E(M)}(B)$, the normalizer of a three-cycle subgroup in S_4. It is S_3, with quotient C_2 by B. Its involution inverts B, so its image in $\overline{N}$ is a reflection. It is outside K, since the stabilizer in K is trivial. Every external reflection has the form $x^{2a+1}s$; changing the initial reflection generator permits $j = xs$. Conjugation by j sends r to r^{-1} and s to rs. Since j fixes the identity point M in the regular set, these conjugation formulas are exactly its action on the coordinates. □

2 Different parabolic types

The common overgroup calculation used in the source audit has the following interface: if $H \leq G$ properly contains B and its action on H/B is two-transitive, then H is C_6, S_3 or A_4. For this particular B the list can also be checked directly. Writing $d = [H : B]$, transitivity of B on the other $d - 1$ cosets gives $d - 1 \mid 3$, so $d = 2$ or 4. At degree two H has order six and is cyclic or symmetric. At degree four the kernel, being contained in B, has order one or three. A kernel of order three would leave a group of order four acting two-transitively on four points, impossible. Thus the action is faithful and its image is the unique subgroup A_4 of order twelve in S_4. This proves the interface without an additional finite search.

Theorem 2. *The configurations with parabolics $M \cong A_4$ and $H \cong C_6$ exist exactly when m is even; then there is one conjugacy class and one E-class. With $H \cong S_3$, there are m conjugacy classes and $\lceil m/2 \rceil$ classes under E.*

Proof. A cyclic overgroup C_6 must lie in the unique torus centralizing B. It exists exactly when $6 \mid n$, equivalently m even, and is then unique. Fixing M and its subgroup B loses no configuration because both choices form a single chain class. This proves the first count.

A subgroup S_3 containing B normalizes B and is the inverse image of a reflection subgroup of order two in K. Conversely each reflection $r^k s$, with $k \in \mathbb{Z}/m\mathbb{Z}$, gives one such S_3. Indeed its inverse image has order six and inverts B. There are exactly m possibilities. The stabilizer of the chain (M, B) inside G is B, which fixes every one of these overgroups. Thus all m give distinct conjugacy classes of configurations. For the E-classes we further quotient by j, whose action is $k \mapsto 1 - k$. This involution has one fixed point for odd m and none for even m, proving the count $\lceil m/2 \rceil$.

In both types, the intersection with M is exactly B: B is maximal in M and M cannot lie in a group of order six. Neither group of order six is a subgroup of A_4, so M and H generate G by maximality of M. Their local actions are respectively the natural action of A_4 of degree four and an action of degree two. This verifies the rank two geometry and local two-transitivity conditions, rather than just counting abstract pairs. □

3 Equal parabolic types and inversion

Lemma 3. *If a group K acts regularly on a set, its orbits on ordered pairs of distinct points are parametrized by $K \setminus \{1\}$. Its orbits on unordered pairs are parametrized by the relation $k \sim k^{-1}$ on that same set.*

Proof. Using the regular coordinates, a pair (a, b) has invariant $a^{-1}b$. Translate its first entry to the identity to see that this invariant is complete. Reversing the pair replaces the invariant by its inverse. These are exactly the further identifications when order is forgotten. □

Theorem 4. *Under the source's unordered convention, the configurations with two parabolics A_4 meeting in B have*

$$m + \left\lfloor \frac{m}{2} \right\rfloor \quad \textit{conjugacy classes, and} \quad m \quad \textit{classes under } E.$$

Proof. Two distinct members of $\mathcal{A}$ meet in exactly B, since B is maximal in each A_4. They generate G, and both local actions have degree four and are two-transitive. Conversely every configuration of the specified type occurs this way. Any conjugacy between two such pairs normalizes their intersection B. The conjugacy problem is therefore precisely the regular action of K on unordered pairs in $\mathcal{A}$.

All m reflections in K are self-inverse. Among the nonidentity rotations, inversion has $\lfloor m/2 \rfloor$ orbits. Lemmas 1 and 3 give the asserted inner count. For E, the additional relation is $k \sim \theta(k)$. On rotations this already is inversion. On reflections inversion is trivial, whereas θ is $k \mapsto 1 - k$ on their exponents. Thus the number of classes under E is

$$\left\lfloor \frac{m}{2} \right\rfloor + \left\lceil \frac{m}{2} \right\rceil = m.$$

This also shows exactly why the source's full automorphism count survives the correction to its conjugacy count. □

4 The two source passages

Table 2 of the published Part II article has seven rows. Its first row is Theorem 2 for C_6; combining m even with the maximal-A_4 congruences gives exactly $p \equiv 13, 37, 83, 107 \pmod{120}$. Rows 2–5 are the S_3 case, separated by the sign of $p - \epsilon$ and the parity of m. All five rows retain their two class counts.

Rows 6–7 concern two A_4 parabolics. They print the inner count $(p-\epsilon)/3-1 = 2m-1$, which counts ordered pairs in the regular set. Section 2 of the published source defines configurations using unordered sets of parabolics. Their correct inner count under that definition is $m + \lfloor m/2 \rfloor$. The printed full automorphism count $(p-\epsilon)/6 = m$ is recovered. For example, the inner count at $p = 37$ is nine instead of eleven, and at $p = 43$ it is ten instead of thirteen. These are consequences of the uniform proof, not numerical searches.

The fixed appendix repeats the same calculation in Proposition 16, Subcase 3, on its physical page 4. It subtracts one from the regular set size but does not then divide by inversion. The same correction therefore disposes of that count in the second original. Its Subcases 1–2, physical pages 2–4, have the recovered different-type counts proved above. The exceptional case $p = 5$ has $m = 1$ and both counts equal one, so the correction does not alter that row's value.

The original configurations and the classical subgroup premises are attributed to the source and its cited subgroup classifications. The common normalizer description, explicit reflection action and inversion quotient are the derivation used here. This component does not complete either original, add a principal refutation, claim a new open-conjecture answer, claim Lean verification or measure compression.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* PSL$(2, q)$*: part II*, Ars Mathematica Contemporanea **6** (2013), 365–388, Section 2, Table 2 and Proposition 5.6.

[2] The same authors, *Part II (Appendix)*, retained 21-page edition, Proposition 16, physical pp. 2–4.

Exceptional parabolics from their common normalizers

Human Compression Project

Abstract

A fixed small intersection has at most two exceptional overgroups of the required type. Counting those overgroups and then forgetting their order recovers the exceptional-family rows of the Part II classification. Dihedral second parabolics are uniquely specified by their rotations; mixed A_5, S_4 pairs are specified by an A_4 normalizer orbit. The same arguments cover both conjugacy classes and their fusion, without a numerical group census.

1 Scope and the two-class argument

Let $G = \mathrm{PSL}(2, q)$ with q odd. For $M \cong S_4$ assume $q = p$ is prime and $q \equiv \pm 1 \pmod 8$. For $M \cong A_5$ assume either $q = p \equiv \pm 1 \pmod 5$ with p odd prime, or $q = p^2 \equiv -1 \pmod 5$ with p odd prime. In these cases M is maximal, has two G-classes, and these classes are fused by $E = \mathrm{PGL}(2, q)$. These are the parameter ranges of the fixed Part II source [1]. The classical subgroup facts, as well as the small subgroup lattices of A_5 and S_4, are retained premises.

The common two-transitive overgroup lemma is a separate explicit interface for exhaustivity. It leaves the following possibilities:

M	B	possible other parabolic
A_5	D_{10}	D_{20}, D_{30}, A_5
A_5	A_4	S_4, A_5
S_4	D_6	D_{12}, D_{18}, S_4
S_4	D_8	D_{16}, D_{24}, S_4
S_4	A_4	A_5

A second S_4 over the normal subgroup A_4 would have to equal its unique ambient normalizer and is not a distinct parabolic. The lists are subject to the actual ambient divisibility conditions. The common lemma is proved in the separately reviewed manuscript *Two-transitive coset actions in dimension two*. All counting arguments below are independent of that exhaustivity step.

Lemma 1. *Suppose M has two ambient conjugacy classes exchanged by E, contains a single internal class of subgroups of type B, and $\mathrm{N}_M(B) = B$. Assume M is self-normalizing and B is maximal in M. Suppose also that the B subgroups form one E-class.*

1. *If B has two G-classes and $[\mathrm{N}_G(B) : B] = 2$, then there are exactly two copies of M above each B, both in the same G-class.*
2. *If B has one G-class and $\mathrm{N}_G(B) = B$, then there are exactly two copies of M above it, one in each G-class.*

Consequently unordered pairs of distinct copies of M meeting in B have respectively two or one conjugacy classes, and one class under E in either case.

Proof. For a fixed ambient class of M containing B, its members above B form a transitive $\mathrm{N}_G(B)$-set, by adjusting a transporter inside the target M. Their stabilizer is B. In the first case, each M class contains only one B class. The outer interchange of the two classes of each type forces the two M classes to contain different B classes. Hence the fixed B has exactly two overgroups in one class and none in the other. In the second case, each of the two M classes contributes exactly one member.

In both cases there is a unique unordered pair above B, and its intersection is exactly B by maximality of B in M. The intersection recovers B from the pair. Thus conjugacy and E classes of unordered pairs are exactly those of B itself. Distinct maximal parabolics generate G. Their prescribed local actions are two-transitive, so these pairs give the required geometries. □

2 Dihedral intersections

Let $B = D_{2d}$ with $d = 3, 4$ or 5. Its characteristic rotation subgroup lies in a unique torus of G, of order $n = (q - \epsilon)/2$, where $\epsilon = \pm 1$ and $d \mid n$. Write $u = n/d$. In the torus normalizer of E choose

$$\langle R, S \mid R^{2n} = S^2 = 1,\ SRS = R^{-1} \rangle, \qquad \mathrm{N}_G(C_d) = \langle R^2, S \rangle, \qquad B = \langle R^{2u}, S \rangle.$$

All such B form one E-class: the rotations are fixed by the torus and the possible reflections in G differ by even powers of R, which are conjugate under its full cyclic group. A rotation R^k normalizes B exactly when $u \mid k$. Therefore $\mathrm{N}_E(B) = \langle R^u, S \rangle \cong D_{4d}$. Its intersection with G is all of D_{4d} if u is even, and is B if u is odd. The E-class splits into two G-classes exactly when its normalizer lies in G, hence exactly when u is even.

Theorem 2. *Let (M, B) be one of*

$$(A_5, D_{10}), \qquad (S_4, D_6), \qquad (S_4, D_8).$$

The equal-type configurations $(M, M; B)$ have two conjugacy classes if $u = (q - \epsilon)/(2d)$ is even and one if it is odd. Their E-class count is one. For $k = 2$ or 3, configurations $(M, D_{2kd}; B)$ exist exactly when $kd \mid n$, and then have two conjugacy classes and one E-class.

Proof. Each indicated B is maximal and self-normalizing in M, and forms one internal class there. The equal-type counts follow directly from Lemma 1 and the normalizer calculation above.

A dihedral overgroup $H = D_{2kd}$ must normalize the same rotation torus: its cyclic rotation subgroup contains C_d. If $kd \mid n$, that torus has a unique subgroup C_{kd}, and $H = \langle C_{kd}, B \rangle$ is the unique overgroup of the required type containing the fixed B. Conversely its existence requires that divisibility. For each ambient class of M there is one class of chains (M, B), and the unique H over B then gives one configuration. The two classes of M give two conjugacy classes, interchanged by E.

The intersection $M \cap H$ equals B: otherwise maximality of B in M would place M inside a dihedral group, impossible. Maximality of M then gives generation of G. The action of D_{4d} on cosets of D_{2d} has degree two; that of D_{6d} has degree three with quotient S_3. These are two-transitive. The M-side actions are their standard small actions. □

The parity and divisibility tests recover Table 1 rows 5–8 and 12–15, and Table 3 rows 1–8, with the parameter qualifications below. In Table 3, $6 \mid n$, $9 \mid n$, $8 \mid n$ and $12 \mid n$ are exactly the printed congruences for rows 1,2,5,6. The full graph groups for the S_4, S_4, D_8 and S_4, D_{16}, D_8 cases are proved in separate accepted manuscripts; these counts do not substitute for those group-identification arguments.

3 The intersection A_4

The classical subgroup normalizers give

$$\mathrm{N}_G(A_4) = \begin{cases} S_4, & q \equiv \pm 1 \pmod 8, \\ A_4, & q \equiv \pm 3 \pmod 8. \end{cases}$$

In the first case there are two classes of A_4, and in the second there is one; they form one E-class. Within A_5 the subgroups A_4 form one class and are maximal and self-normalizing.

Theorem 3. *The configurations $(A_5, A_5; A_4)$ have two conjugacy classes when $q \equiv \pm 1 \pmod 8$ and one when $q \equiv \pm 3 \pmod 8$. In both cases they have one E-class. Configurations $(A_5, S_4; A_4)$ exist in the first case only, and have two conjugacy classes and one E-class.*

Proof. The equal-type assertion is Lemma 1. For the mixed pair, a fixed $B = A_4$ has the unique S_4 overgroup $N = \mathrm{N}_G(B)$. The two A_5 classes contain different A_4 classes, since the outer coset interchanges each pair of classes. Thus the fixed B belongs to only one class of A_5. Its members above B form a transitive N-set with stabilizer B, so there are two of them, interchanged by N. Hence there is one mixed configuration for each of the two B classes, and the outer automorphism interchanges them. The intersection is B; local indices are five and two. At least the A_5 parabolic is maximal, so the graph is connected. For $q \equiv \pm 3 \pmod 8$, the ambient group contains no S_4, so the mixed configuration does not exist. □

These statements recover Table 1 rows 9–11 and 16–18, and Table 3 row 9. The mixed $q = 9$ case is also Table 6 row 2; its identical normalizer argument is used in the subfield manuscript. For these families there is already one class under E, so adjoining field automorphisms cannot change the full automorphism count.

4 Empty parameter rows and reuse

In the square-field part of Table 1, the hypotheses impose $q = p^2 \equiv -1 \pmod 5$ with p odd. Every odd square is one modulo eight; combining this with $q \equiv 4 \pmod 5$ gives

$$q \equiv 9 \pmod{40}.$$

Consequently rows 12–14 and 18 have no admissible parameters: row 12 requires $q \equiv -1 \pmod{20}$; row 14 requires $(q+1)/10$ even; row 18 requires $q \equiv 29$ or $19 \pmod{40}$. Row 13 requires $q \equiv -1 \pmod{30}$, impossible since a square is never two modulo three. These are empty conditional branches, not numerical counterexamples. Rows 15–17 are the actual square-field configurations. In rows 16–17 the printed alternative $q \equiv -1 \pmod{40}$ is likewise empty, leaving $q \equiv 9$.

The remaining cases of Table 1 are already accounted for elsewhere: rows 2–4 use the accepted scalar-orbit proof in characteristic two; row 1 uses the unique D_{30} overgroup in the subfield manuscript, and repeats Table 4 row 5. That row's prime-exponent condition reduces to $q = 16$. The characteristic-five odd-degree case supplies no geometry, by the subfield uniqueness and exclusion proof. Thus no A5-family branch is silently omitted.

For source attribution, these are the configurations and numerical claims of L063 Propositions 5.5 and 5.10, Table 1 and Table 3. The retained L064 appendix gives the S4 counting argument in Proposition 20; its proof is recovered by the same formulas. The source's advertised A5-count

appendix pages are not present in that fixed edition, so the published A5-family proof is joined directly. The classical subgroup premises and the original constructions remain attributed. The common normalizer proof and explicit unordered-pair argument are our derivation. No additional principal refutation, conjecture answer, Lean verification or compression measurement is claimed. Whole-original completion still requires the source ledgers and the other shared proof components.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* PSL$(2, q)$*: part II*, Ars Mathematica Contemporanea **6** (2013), 365–388, Table 1, Table 3 and Propositions 5.5, 5.10.

[2] The same authors, *Part II (Appendix)*, retained 21-page edition, Proposition 20.

Subfield parabolics and affine overgroups

Human Compression Project

Abstract

The stabilizer of a subfield line determines its projective overgroup, while its affine overgroups are controlled by invariant vector spaces. In even characteristic, one-dimensional quotients supply the additional geometries omitted by the source. For an odd extension in odd characteristic the available scalar group is too small to do so. The quadratic projective case has exactly two conjugacy classes of affine configurations, interchanged by an outer automorphism. We give these arguments in common coordinates and then treat the small exceptions separately.

1 Premises and the representation interface

Let $K = \mathbb{F}_s \subset F = \mathbb{F}_q$ and let $G = \mathrm{PSL}(2, F)$ act on $F \cup \{\infty\}$. For an additive subgroup $V \leq F$, write $U_V = \{x \mapsto x + v : v \in V\}$. Let C_D denote multiplication by elements of a subgroup $D \leq F^*$. The scalars present in the stabilizer of ∞ in G form

$$A_F = \begin{cases} F^*, & \mathrm{char}\, F = 2, \\ (F^*)^2, & \mathrm{char}\, F \text{ odd}. \end{cases}$$

All products $U_V : C_D$ below use these literal transformations.

We retain the classical Dickson subgroup classification, its subfield embedding and conjugacy statements, and its small exceptional group normalizers. In particular, odd-degree $\mathrm{PSL}(2, K)$ subgroups form one conjugacy class, as do the even-characteristic subfield groups. The two classes of $\mathrm{PGL}(2, K)$ in $\mathrm{PSL}(2, \mathbb{F}_{s^2})$ are interchanged by $\mathrm{PGL}(2, \mathbb{F}_{s^2})$. We do not infer configuration counts from these subgroup counts alone.

The following is the explicit interface with the separate common representation proof. It is not an import of the source's unpublished Proposition 4.6: for a subgroup of projective type, a two-transitive coset action has its natural Borel stabilizer, or one of the following exceptional stabilizers:

group	additional point stabilizer
$\mathrm{PSL}(2,4) \cong \mathrm{PSL}(2,5)$	D_{10} or A_4, as applicable
$\mathrm{PSL}(2,7)$	S_4
$\mathrm{PSL}(2,9),\ \mathrm{PSL}(2,11)$	A_5
$\mathrm{PGL}(2,t)$ (t odd)	$\mathrm{PSL}(2,t)$ (degree two)
$\mathrm{PGL}(2,5)$	S_4

The familiar $\mathrm{PSL}(2,2) = S_3$, $\mathrm{PSL}(2,3) = A_4$ and $\mathrm{PGL}(2,3) = S_4$ actions are included with their elementary kernels. The same common proof supplies the small exceptional overgroup lists when the point stabilizer is A_4, S_4, A_5 or a small dihedral group. This interface is proved in the separately reviewed manuscript *Two-transitive coset actions in dimension two.* The coordinate and uniqueness lemmas below are independent of it.

2 A fixed Borel has a unique projective overgroup

Lemma 1. *Suppose $|K| = s > 2$ and $D \leq K^*$ is nontrivial. For $B = U_K : C_D \leq G$,*

$$\mathrm{N}_G(B) = U_K : C_{K^* \cap A_F}.$$

In particular $\mathrm{N}_G(B) = B$ in each of these cases:

1. *$s \geq 4$ is even, $D = K^*$;*
2. *$s \geq 5$ is odd, $[F : K]$ is odd, $D = (K^*)^2$;*
3. *$s \geq 3$ is odd, $[F : K] = 2$, $D = K^*$.*

Proof. The unique Sylow subgroup U_K for the characteristic of F is characteristic in B. It has the unique common fixed point ∞, so a normalizer of B has the form $x \mapsto ax + b$ with $a \in A_F$. Normalizing U_K gives $aK = K$, hence $a \in K^*$. Conjugating a scalar $c \neq 1$ in D gives translation term $(1 - c)b$. This belongs to K exactly when $b \in K$. Conversely these conditions plainly normalize B, proving the formula. For an odd-degree extension, an element of K^* is a square in F exactly when it is a square in K; for a quadratic extension every element of K^* is a square in F. The assertions follow. □

Lemma 2. *In the three cases of Lemma 1, respectively put $M = \mathrm{PSL}(2, K)$, $M = \mathrm{PSL}(2, K)$ and $M = \mathrm{PGL}(2, K)$. Assume the extension degree is prime, so that the specified M is maximal in G. Then exactly one conjugate subfield group of this type contains the fixed B as a natural Borel.*

Proof. The Borel subgroups inside M form one M-class and are self-normalizing in M. For any ambient conjugacy class of copies of M that contains B, their set above B is a transitive $\mathrm{N}_G(B)$-set: adjust a transporter inside the target M using its internal Borel conjugacy. Its stabilizer is B, because M is maximal and self-normalizing. Lemma 1 makes this set a singleton.

In the first two cases there is only one ambient class of M. In the third case there are two. The Borel subgroups have two ambient classes as well, which can be seen directly. After moving their unique fixed point to ∞, their radicals are the subfield lines aK. Conjugation multiplies a by an element of A_F. As $K^* \subset A_F$, the quotient $F^*/(A_F K^*)$ has order two. The two classes are interchanged by any nonsquare scalar in $\mathrm{PGL}(2, F)$. That same outer coset interchanges the two classes of M. Since one class of M contains only one class of Borels, the two classes of M must contain different Borel classes. Thus only one class of M can contain the fixed B, completing the proof. □

Here the radical description is literal: its scalar complement K^* forces an additive radical of order s to be a one-dimensional K-space. Any translated scalar complement can be made linear by an affine translation. The same argument gives one Borel class in the first two cases: $A_F K^* = F^*$ for an odd-degree odd-characteristic extension, and $A_F = F^*$ in even characteristic.

3 All affine covers of the fixed Borel

Lemma 3. *In any of the three cases above, let H be an affine subgroup of G containing B properly, with B maximal in H. If $D = K^*$, then the two-transitive possibilities are exactly*

$$H = U_V : C_{K^*}, \qquad K \subset V \subset F, \qquad \dim_K V = 2.$$

If $D = (K^*)^2$ and $s \geq 5$ is odd in an odd-degree extension, there is no such two-transitive affine overgroup.

Proof. An affine group containing U_K must fix ∞, since that is its only possible common fixed point. Write U_V for the translation kernel of H. The group D generates the field K: this is immediate for K^*; for $(K^*)^2$ its order is $(s-1)/2 > \sqrt{s}-1$, larger than the multiplicative group of any proper subfield. Thus V is a K-space containing K.

We first see that the scalar part of H can be taken literally. If $x \mapsto ax+b$ belongs to H, commute it with a fixed scalar $c \neq 1$ in D. The resulting translation is a nonzero K-multiple of b, so $b \in V$. Removing this translation gives the scalar a itself in H. Therefore $H = U_V : C_{D'}$ for some $D \leq D' \leq A_F$ stabilizing V. If $V = K$, then $D' \leq K^* \cap A_F = D$, contrary to properness. If $V > K$, the intermediate subgroup $U_V : C_D$ properly contains B, so maximality forces $D' = D$.

Identify the cosets of B in H with the additive quotient V/K. Translations act regularly and the point stabilizer induces exactly the scalar group D. Hence two-transitivity is equivalent to transitivity of D on the nonzero vectors of V/K. If $D = K^*$, this holds exactly when $\dim_K(V/K) = 1$: the orbit has size at most $s-1$, and a one-dimensional quotient has exactly $s-1$ nonzero vectors. For $D = (K^*)^2$, even a one-dimensional quotient has more nonzero vectors than $|D|$, so transitivity is impossible. □

The scalar case with $D = K^*$ is the previously accepted affine-plane construction. Its complete parametrization and Frobenius action are reused from *Scalar orbits and subfield configurations*; they are not a new acquisition here. Lemma 3 explains why no other affine radical or enlarged complement can occur.

Theorem 4. *Assume the representation interface in Section 1.*

1. *For even $s \geq 4$ and prime $[F : K] = m$, the configurations with $M = \mathrm{PSL}(2, K)$ and natural Borel intersection are precisely $\{M, U_V : C_{K^*}\}$ with $\dim_K V = 2$ and $K \subset V$. They have $(s^{m-1}-1)/(s-1)$ conjugacy classes. Their full semilinear classes are the Frobenius orbits on the same projective parameter set. At $m = 2$ both counts are one.*

2. *For odd $s \geq 5$ and odd prime extension degree, there is no configuration with $M = \mathrm{PSL}(2, K)$ and its natural Borel intersection.*

3. *For odd $s \geq 3$ and $F = \mathbb{F}_{s^2}$, there are exactly two conjugacy classes and one automorphism class of configurations with $M = \mathrm{PGL}(2, K)$ and natural Borel intersection. Their other parabolic is $U_F : C_{K^*}$.*

Proof. The representation interface and the characteristic restrict a non-affine overgroup of the given Borel to the same projective type as M. Indeed, the natural Borel stabilizer determines its field from its characteristic part and scalar order. A cyclic group cannot contain B. A dihedral candidate can occur only for the small coincidences $B = D_{10}$ in characteristic five or $B = D_6$ in characteristic three; a proper dihedral overgroup would require a cyclic element with mixed characteristic order, or order nine in characteristic three, neither of which occurs in a projective group in that characteristic. The exceptional groups A_4, S_4, A_5 are checked by their orders and the small-overgroup interface. The small coincidence $U_4 : C_3 = A_4$ gives $A_5 = \mathrm{PSL}(2, 4)$ in characteristic two; the coincidence $U_5 : C_2 = D_{10}$ gives $A_5 = \mathrm{PSL}(2, 5)$ in characteristic five. They introduce no new projective overgroup. Lemma 2 excludes a second such copy. The remaining possibilities are exactly those in Lemma 3.

For the first case, the accepted scalar-orbit proof gives the stated counts and parameter set. In the second case there is no affine possibility. In the third case the only two-dimensional K-space is F itself. Each fixed Borel therefore has exactly one projective and one affine overgroup, so

its two ambient classes give exactly two configuration classes. A nonsquare scalar exchanges the two, proving the single automorphism class. All intersections are exactly B; maximality of M gives generation of G, and the affine action on F/K supplies local two-transitivity. □

4 Odd subfields and the exceptional stabilizers

Here a subgroup defined over K means that every projective element lies in $\mathrm{PGL}(2, K)$ and is individually fixed by the Frobenius of F/K. Mere stability of the subgroup as a set is insufficient.

Lemma 5. *Let $K \subset F$ be finite fields. Suppose $L_1, L_2 \leq \mathrm{PGL}(2, K)$ have trivial centralizer in $\mathrm{PGL}(2, F)$. Every element of $\mathrm{PGL}(2, F)$ conjugating L_1 to L_2 belongs to $\mathrm{PGL}(2, K)$. If the extension degree is odd, two such subgroups of $\mathrm{PSL}(2, K)$ that are not conjugate there cannot become conjugate in $\mathrm{PSL}(2, F)$.*

Proof. Let $gL_1g^{-1} = L_2$ and let σ be the Frobenius generating $\mathrm{Gal}(F/K)$. For every $h \in L_1$, both h and ghg^{-1} are fixed by σ. Thus $g^{-1}\sigma(g)$ centralizes L_1 and is the identity. A projective matrix fixed by σ has a representative over K: normalize one nonzero entry to one, after which all entry ratios are fixed. Hence $g \in \mathrm{PGL}(2, K)$. Finally $\mathrm{PGL}(2, K) \cap \mathrm{PSL}(2, F) = \mathrm{PSL}(2, K)$ for odd extension degree, by the square criterion used in Lemma 1. □

Lemma 6. *Let M be the standard $\mathrm{PSL}(2, K)$ or $\mathrm{PGL}(2, K)$ in $\mathrm{PGL}(2, F)$. A subgroup $B \leq M$ with trivial projective centralizer is contained in no other $\mathrm{PGL}(2, F)$-conjugate of M.*

Proof. Suppose $B \leq gMg^{-1}$. Then $g^{-1}Bg$ and B are both subgroups of $\mathrm{PGL}(2, K)$, and their centralizers are trivial. Lemma 5 gives $g \in \mathrm{PGL}(2, K)$. That group normalizes both possible choices of M, so $gMg^{-1} = M$. □

In the odd-characteristic fields used in this section, the groups A_4, S_4, A_5 have trivial projective centralizer. Here is an elementary check. In characteristic three choose two distinct cyclic subgroups of order three: their unipotent centralizers have different fixed points and intersect trivially. Indeed, two distinct such subgroups with the same fixed point would commute and generate a group of order nine, which cannot lie in any of the three groups under consideration. In other characteristics the centralizer of each such subgroup is its torus. An element centralizing both must lie in both tori. A nonidentity noninvolution has its own unique torus centralizer; an involution has a dihedral centralizer whose odd-order subgroups lie in its torus. In either case both subgroups of order three would be the same unique subgroup of order three in that torus, a contradiction.

Theorem 7. *Assume the common representation interface. For $q = s^m$ odd with m an odd prime and maximal $M = \mathrm{PSL}(2, s)$, the only configurations are those with $s = 3$ and two A_4 parabolics meeting in C_3. Their counts are the corrected counts in the accepted scalar-orbit proof. In particular no additional configurations arise from the exceptional A_4, S_4, A_5 stabilizers at $s = 5, 7, 9, 11$.*

Proof. For $s \geq 5$ the natural Borel case is excluded by Theorem 4. The exceptional point stabilizer is A_4 at $s = 5$, S_4 at $s = 7$, and A_5 at $s = 9, 11$. The small overgroup interface in the actual characteristic leaves only another copy of the same M as a two-transitive overgroup. All copies of this subfield group are conjugate under $\mathrm{PGL}(2, F)$, by the retained subfield classification. The exceptional stabilizers have trivial projective centralizer, as just proved. Lemma 6 therefore gives

a unique M containing the fixed stabilizer, ruling out a second parabolic. This argument also explains why simply fusing internal subgroup classes would not be an adequate replacement for a literal overgroup proof.

At $s = 3$, $B = C_3$ is unipotent. Its normalizer in G is U_F: an affine multiplier preserving the translation line $\mathbb{F}_3$ is ± 1, and -1 is a nonsquare in F because m is odd. The elementary degree argument for a subgroup with point stabilizer C_3 gives C_6, S_3, A_4. The first two would normalize C_3 and contain an involution, impossible in U_F. Only the A_4 overgroups remain. Their complete conjugacy and Frobenius-orbit counts, with unordered interchange, were proved in the accepted scalar-orbit component. □

5 The small even-field cases

Theorem 8. *The case $M = \mathrm{PSL}(2,2)$ in $G = \mathrm{PSL}(2,4)$ gives exactly one conjugacy and one automorphism class of each of the following types:*

$$(S_3, S_3; C_2), \qquad (S_3, C_2^2; C_2), \qquad (S_3, A_4; C_3).$$

For $M = \mathrm{PSL}(2,4)$ in $G = \mathrm{PSL}(2,4^m)$ with m prime, the exceptional intersection D_{10} gives the single type $(A_5, D_{30}; D_{10})$, with one class in both senses, exactly when $15 \mid 4^m \pm 1$. This condition is equivalent here to $m = 2$.

Proof. Use the natural A_5 model for $\mathrm{PSL}(2,4)$. A subgroup S_3 is the normalizer of a three-cycle subgroup; its involutions are products of two transpositions. It is maximal in A_5. There are ten such subgroups, fifteen involutions, five point stabilizers A_4, and five subgroups C_2^2; each latter group is the centralizer of any of its nonidentity elements. These counts also follow directly by choosing the supports of cycles in five letters.

A point stabilizer C_2 in a two-transitive action has degree two or three, since $d - 1 \mid 2$. Thus its overgroup has order four or six. The order-four possibility in A_5 is C_2^2, uniquely specified by the involution. The order-six possibility is S_3. Each involution belongs to exactly two S_3 subgroups: count $10 \cdot 3 = 30$ incidences and use transitivity on the fifteen involutions. Their unordered pair is unique, proving the first configuration count. For the different-type pair S_3, C_2^2, all chains consisting of S_3 and one of its involutions are conjugate, proving the second count.

For the intersection C_3, the small degree argument leaves C_6, S_3, A_4. There is no C_6 in A_5, and the normalizer S_3 of the fixed C_3 is its unique order-six overgroup. Exactly two point stabilizers A_4 contain this C_3, corresponding to its two fixed letters. Its S_3 normalizer interchanges these two letters. Hence there is exactly one configuration of the third type. Each asserted intersection and each local action follows from the indicated elementary subgroup lattices. Maximality of the S_3 parabolic gives generation and connectedness.

Now let $q = 4^m$. The two-transitive overgroup interface for D_{10} leaves A_5, D_{20}, D_{30}. The cyclic torus containing C_5 has odd order $q \pm 1$. The normalizer of the specified D_{10} is D_{10} itself: a torus element normalizing it has square in C_5, and squaring is invertible on the odd-order quotient torus. There is one class of A_5 subfield groups and one internal class of D_{10}, so the transporter argument gives a unique A_5 above the fixed intersection. An overgroup D_{20} is impossible, since it would contain an element of order ten in characteristic two.

A D_{30} overgroup exists precisely when the same torus contains C_{15}, and is then uniquely $\langle C_{15}, D_{10} \rangle$. The D_{10} subgroups form one ambient class, proving both counts. Their local indices are six in A_5 and three in D_{30}; the corresponding natural actions are two-transitive. At $m = 2$ the condition holds because $4^2 - 1 = 15$. If m is odd, five divides $4^m + 1$, while three divides $4^m - 1$, so no torus order is divisible by fifteen. This proves the last assertion. □

6 The remaining quadratic projective intersections

Theorem 9. *Let s be odd and $G = \mathrm{PSL}(2, s^2)$, with maximal parabolic $M = \mathrm{PGL}(2, s)$. An intersection $\mathrm{PSL}(2, s)$ gives no second parabolic unless $s = 3$, in which case the other parabolic is A_5. That exceptional configuration has two conjugacy classes and one automorphism class. The exceptional S_4 intersection at $s = 5$ gives no configuration with a second parabolic.*

Proof. Put $L = \mathrm{PSL}(2, s)$. Its ambient normalizer is M, by the classical subfield normalizer formula. Any other $\mathrm{PGL}(2, s)$ overgroup would have L as its characteristic subgroup of index two, and would therefore lie in $\mathrm{N}_G(L) = M$. Hence it equals M. The representation and small-overgroup interface leaves only A_5 as an additional possibility, and only when $s = 3$, so that $L = A_4$. For clarity, an overgroup of larger projective field cannot create a hidden option: the subfield classification forces its field to contain $\mathbb{F}_s$. Inside $\mathbb{F}_{s^2}$ the only larger possibility is the entire ambient group. Its index over L is $s(s^2 + 1)$, which exceeds $|L| + 1$ and violates the two-transitivity order bound.

For $s = 3$, the group $G = A_6$ has two classes of A_4, with normalizer S_4, and two classes of A_5. Each A_5 has one internal class of A_4. An outer automorphism interchanges the two classes of each type; hence the two A_5 classes contain different A_4 classes. Fix $L = A_4$. The members of its one applicable A_5 class containing it form a transitive $\mathrm{N}_G(L) = S_4$-set with stabilizer L: the A_5 is self-normalizing and its intersection with S_4 is precisely L. Thus there are two such A_5 overgroups, exchanged by the fixed S_4 parabolic. This gives one configuration class for each of the two A_4 classes, and therefore two in all. The outer automorphism fuses them, giving one automorphism class. The local indices are two and five, with the standard two-transitive actions.

At $s = 5$, the subgroup S_4 is the exceptional point stabilizer in $M \cong S_5$. The common overgroup interface leaves only $\mathrm{PGL}(2, 5)$. Its S_4 stabilizer has trivial projective centralizer and is defined over $\mathbb{F}_5$. Lemma 6 gives a unique $\mathrm{PGL}(2, 5)$ overgroup, ruling out a second parabolic. □

The $q = 9$ configuration with two S_4 parabolics and intersection D_8 is imported from the separately accepted duad–syntheme proof, which also identifies its full graph group. It is not proved again here.

7 Source claims, corrections and attribution

The original numerical rows are interpreted under their stated maximal-parabolic hypotheses. The results above give the following joins; the common representation interface remains a separately reviewed dependency.

- L063 Lemma 5.11 and Proposition 5.12, and L064 Lemma 21 / Proposition 22: Theorem 4 recovers Table 4 row 1 at $m = 2$ and supplies the missing affine families for other prime degrees. The published restriction to $m = 2$ is false. The fixed appendix's Lemma 21 already lists the affine type without that restriction, so its statement is distinguished from the false exhaustivity conclusion of Proposition 22.

- Table 4 rows 2–4 are Theorem 8 at $q = 4$. Row 5 repeats Table 1 row 1 and is the D_{10} part of that theorem. The natural-Borel subcase at $s = 4$ also reuses the accepted Table 1 rows 2–4 scalar-orbit counts.

- L063 Lemmas 5.13–5.15 / Proposition 5.16, and L064 Lemmas 23–25 / Proposition 26: Theorem 7 proves the remaining exclusion arguments in their Section 5.4.2 context of odd

prime extension degree. The two surviving Table 5 rows retain the accepted correction for unordered inner counts. The statement of Lemma 5.13 is read with that surrounding odd-degree hypothesis.

- L063 Lemma 5.18 / Proposition 5.21 and L064 Lemma 28 / Proposition 30: Table 6 row 1 is Theorem 4(3). Theorem 9 gives row 2 and the nonexistence conclusions above $\mathrm{PSL}(2, s)$ and, at $q = 25$, above S_4. These also cover L063 Lemmas 5.19–5.20 and L064 Lemma 29. The fixed appendix's Proposition 30 does not include a separate $q = 25/S_4$ subcase; that published-source obligation is proved explicitly here.

  The proof of Lemma 24 in the fixed appendix also states that $\mathrm{PSL}(2, 3^{2m})$ contains no $\mathrm{PGL}(2, 3^w)$ subgroup. That auxiliary assertion is false: $w = 1$ and $w = m$ are allowed subfield embeddings. It is not used here. In the latter case, 5 does not divide $|\mathrm{PGL}(2, 3^m)|$ for odd m, so it cannot contain A_5; the full exclusion is supplied by the explicit representation interface.

- Table 6 row 3, L063 Lemma 5.17 and L064 Lemma 27 are joined to the accepted $q = 9$ component, together with the small overgroup list. Their full graph properties are already covered separately.

The original family constructions and classical subgroup theory retain their attribution. The coordinate affine-cover argument and the Galois-transporter descent argument are the common proofs used here; the earlier scalar-orbit and exceptional-geometry manuscripts are explicitly reused. This manuscript does not claim a new principal refutation, a new answer to an open conjecture, Lean verification or a compression measurement. Whole-original disposition requires the remaining source components and their exact claim ledgers.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* $\mathrm{PSL}(2, q)$*: part II*, Ars Mathematica Contemporanea **6** (2013), 365–388, Sections 5.4–5.5 and Tables 4–6.

[2] The same authors, *Part II (Appendix)*, retained 21-page edition, Lemmas 21–29 and Propositions 22, 26, 30.

Local arc transitivity from small stabilizers

Human Compression Project

Abstract

We determine the local arc lengths of the configurations in Tables 1–6 of Buekenhout, De Saedeleer and Leemans, *On the rank two geometries of the groups* PSL$(2, q)$*: part II.* The proof uses orders of arc stabilizers, kernels of small permutation actions, and one argument for affine subfield covers. Two configurations with stabilizers S_4 require a longer calculation; the same elementary facts about Sylow two-subgroups settle both. These results concern the specified acting group. They do not determine the full automorphism group of an incidence graph.

1 Arcs and stabilizers

Let X be a connected bipartite graph with vertex types $0, 1$, and let G act faithfully, preserve types, and be transitive on edges. Suppose the degrees d_0, d_1 are at least two. An s-arc is a sequence $(x_0, \dots, x_s)$ of adjacent vertices with $x_{i-1} \neq x_{i+1}$. The action is locally s-arc-transitive if G_x is transitive on the s-arcs starting at x, for each vertex x. We assume local two-transitivity, equivalently local 2-arc-transitivity in this setting. Write K_x for the kernel of G_x on $\mathcal{N}(x)$.

Lemma 1. *The number of s-arcs starting at a vertex of type i is*

$$a_i(s) = d_i \prod_{j=1}^{s-1} (d_{i+j \bmod 2} - 1).$$

Local s-arc-transitivity implies $a_i(s) \mid |G_x|$ for both types. Given local s-arc-transitivity, extension to $s+1$ is equivalent to transitivity of $G_{x_0 \cdots x_s}$ on $\mathcal{N}(x_s) \setminus \{x_{s-1}\}$ for each of the two starting types.

Proof. There are d_i choices for the first edge and, at each later step, all neighbors except the preceding vertex are allowed. This proves the product formula. Orbit–stabilizer gives divisibility. To compare two longer arcs, first carry their initial s-arcs to one another; the remaining possible maps are precisely the stabilizer of that initial arc. This proves the last assertion, including its converse. □

In particular, the stabilizer of a 2-arc (x, y, z) is a pointwise two-point stabilizer in the action of G_y on its neighbors. Its order is $|G_y|/[d_y(d_y - 1)]$. This observation uses the middle vertex; confusing it with an endpoint would give the wrong order.

Lemma 2. *If $d_1 = 2$, the graph is locally 3-arc-transitive.*

Proof. For a 2-arc (x, y, z) with y of type 1, the subgroup fixing x inside G_y fixes both neighbors of y, since there are only two. Thus $G_{xyz} = G_{yz}$. Local two-transitivity at z says that this point stabilizer is transitive on $\mathcal{N}(z) \setminus \{y\}$. For the other starting type there is only one possible extension. Apply Lemma 1. □

For a coset graph with parabolics M, L and intersection J, the degrees are $[M : J], [L : J]$. Connectedness is equivalent to $\langle M, L\rangle = G$. The action is locally two-transitive exactly when the actions of M, L on their cosets of J are two-transitive. These facts follow directly by translating an incident pair to the standard pair of cosets. In the applications, the source's retained constructions supply the subgroups and these local actions. The argument below does not use the source's exhaustive classification, which requires a correction elsewhere.

2 Cases settled by orders and prime cycles

We use D_{2m} for the dihedral group of order $2m$. The following table gives all cases with equal parabolic types in which the obstruction to a third arc is immediate.

$M = L$	J	$d_0 = d_1$	$a_0(3)$
A_5	D_{10}	6	150
A_5	A_4	5	80
A_4	C_3	4	36
S_4	S_3	4	36
S_3	C_2	3	12

Each last entry exceeds the parabolic order, so the exact local arc length is two. Reversing the types has no effect on that conclusion.

For degree two at one type, Lemma 2 proves a third arc. The following obstructions prove that the exact length is three:

M	L	J	d_0	$a_1(4) = 2(d_0 - 1)^2$	$\lvert L\rvert$
A_5	D_{20}	D_{10}	6	50	20
A_5	S_4	A_4	5	32	24
A_4	C_6 or S_3	C_3	4	18	6
S_4	D_{12}	S_3	4	18	12
S_3	C_2^2	C_2	3	8	4

Proposition 3. *The configurations (A_5, D_{30}, D_{10}) and (S_4, D_{18}, S_3) have exact local arc length three.*

Proof. Treat the two cases together, with $p = 5$ and $p = 3$, respectively. The degrees are $p + 1, 3$. The action of D_{6p} on its three cosets of D_{2p} has kernel the rotation subgroup C_p and quotient S_3. Indeed, writing $D_{6p} = \langle r, t\rangle$ with $|r| = 3p$, the subgroup is $\langle r^3, t\rangle$, and the quotient by $\langle r^3\rangle$ is the usual degree-three dihedral action.

A 2-arc ending at the first type therefore has stabilizer C_p. Its local action is nontrivial: the degree-six action of A_5 is faithful by simplicity, and the natural degree-four action of S_4 is faithful. On the p available next vertices a nontrivial group of order p is transitive. A 2-arc ending at the second type has stabilizer of order two, since the pointwise two-point stabilizers in these two actions of M have order two. It cannot lie in the kernel C_p, so it exchanges the two available next vertices. This proves local 3-arc-transitivity in both orientations. Finally $a_0(4) = 4p(p + 1) > 2p(p + 1) = |M|$, which excludes a fourth arc. □

3 Affine covers of a subfield point stabilizer

Let $K = \mathbb{F}_s$, with $s \geq 2$, and let V be a vector space of dimension two over K, with a fixed line W. Consider the local configuration

$$M = \mathrm{PGL}(2, K), \qquad L = V \rtimes K^*, \qquad J = W \rtimes K^*.$$

The embedding of J in M is its usual point stabilizer on $\mathbb{P}^1(K)$. In characteristic two $M = \mathrm{PSL}(2, K)$. The argument only uses these embedded local actions, so it also applies when V is a two-dimensional subspace of a larger coefficient field.

Proposition 4. *Every such coset configuration has exact local arc length three.*

Proof. The action of M has degree $s + 1$. The action of L on its cosets of J identifies with the affine action on V/W, of degree s, whose kernel is W. The quotient is $\mathrm{AGL}(1, K)$, which is sharply two-transitive.

For a 2-arc ending at an M-vertex, the middle vertex has stabilizer L, so the arc stabilizer is exactly W. Inside the endpoint group M, this is the translation subgroup of a point stabilizer. It is regular on the s other points. In the reverse orientation the arc stabilizer is a two-point stabilizer of $\mathrm{PGL}(2, K)$, of order $s-1$. It lies in J and has trivial intersection with the translation group W: a nonidentity translation fixes only the point at infinity. It therefore projects onto the full multiplicative group of the affine quotient L/W and is transitive on the $s - 1$ available next vertices. This includes $s = 2$, when that last set has one element.

Thus both orientations extend to three arcs. At an L-vertex the number of 4-arcs is $s^3(s-1)$, whereas $|L| = s^2(s - 1)$. Lemma 1 excludes local 4-arc-transitivity. □

This applies to the $A_5, E_{16} : 3, A_4$ configurations in Table 1, to the even-characteristic subfield row of Table 4 where it exists, and to the $\mathrm{PGL}(2, p^n)$ affine row of Table 6. It also proves the same local value for the missing two-dimensional scalar covers in the corrected even-characteristic classification. Their existence and orbit counts are separate assertions and are not inferred from this local proof.

4 The two-subgroup calculations in S_4

We record the elementary group facts used below. In its action of degree three on partitions into two pairs, S_4 has kernel the normal four-group

$$V = \{1, (12)(34), (13)(24), (14)(23)\}.$$

The three point stabilizers are its Sylow two-subgroups, each isomorphic to D_8. Distinct such subgroups intersect in V. Their three central involutions are the three distinct nonidentity elements of V. These statements follow by taking the stabilizer of $\{\{1, 2\}, \{3, 4\}\}$ and conjugating it to the other two partitions.

In $D_8 = \langle a, b \mid a^4 = b^2 = 1,\ bab = a^{-1} \rangle$ there are exactly two four-groups, $\langle a^2, b \rangle$ and $\langle a^2, ab \rangle$. They intersect in the center $\langle a^2 \rangle$. The unique cyclic subgroup of order four intersects either in that same center. If this D_8 has index two in D_{16}, conjugation by an element outside it interchanges its two four-groups. To see this, write $D_{16} = \langle r, b \rangle$, $|r| = 8$, and $a = r^2$; conjugation by r sends one of the two displayed four-groups to the other.

Proposition 5. *The configuration (S_4, D_{24}, D_8) has exact local arc length three.*

Proof. Both degrees are three. At an S_4-vertex the local kernel is V. At a D_{24}-vertex it is C_4: in the presentation with rotation r of order twelve, the subgroup D_8 is $\langle r^3, b\rangle$, and quotienting by $\langle r^3\rangle$ gives S_3. Since both local quotients are sharply two-transitive of degree three, the stabilizer of a 2-arc is the local kernel of its middle vertex. Inside the endpoint edge stabilizer D_8, neither C_4 nor V is contained in the other. Its nontrivial local image fixes the incoming neighbor and interchanges the other two. This proves the third arc in both orientations.

Choose a 3-arc (a, b, c, d) with a, c of type S_4. The stabilizer of (a, b, c) is C_4 and the stabilizer of the 3-arc is its subgroup of order two. Its generator is a square of an element of order four in $G_a = S_4$, hence is a double transposition in the normal four-group K_a. It fixes every neighbor of a. For the reversed 3-arc (d, c, b, a) it consequently cannot exchange the two possible extensions. Local 4-arc-transitivity fails. □

Proposition 6. *Suppose G is nonabelian simple and is generated by the two parabolics in a configuration (S_4, S_4, D_8). Its exact local arc length is four.*

Proof. Write $V_x = K_x$ at each vertex. If x, y are adjacent and $V_x = V_y$, this nontrivial four-group is normal in both G_x and G_y, and hence in G. This contradicts simplicity. Thus V_x, V_y are the two distinct four-groups in $J_{xy} = G_x \cap G_y$, and

$$V_x \cap V_y = Z(J_{xy}) = \langle z_{xy}\rangle.$$

For a 2-arc (a, b, c), its stabilizer is V_b. Its image on the neighbors of c has order two, since $V_b \cap V_c$ has order two. That image fixes b and interchanges the other two neighbors. The stabilizer of a resulting 3-arc (a, b, c, d) is $V_b \cap V_c = \langle z_{bc}\rangle$.

The subgroups J_{bc} and J_{cd} are different Sylow two-subgroups of $G_c = S_4$, so $z_{bc} \neq z_{cd}$. Both belong to V_c, whereas $V_c \cap V_d = \langle z_{cd}\rangle$. Thus z_{bc} does not belong to the local kernel V_d. It fixes c and exchanges the other two neighbors of d, proving the fourth arc. Finally, $a_i(5) = 48 > 24$, so a fifth arc is impossible for this specified group. □

Proposition 7. *The configuration (S_4, D_{16}, D_8) has exact local arc length seven.*

Proof. Write an alternating path as

$$(v_0, e_0, v_1, e_1, v_2, e_2, v_3, e_3),$$

where the v_i have stabilizer S_4 and the e_i have stabilizer D_{16}. A vertex e_i has just two neighbors. An element fixing e_i and one of them fixes the other; hence the two edge stabilizers there coincide. Write this common group as $J_i \cong D_8$. Let V_i be the local four-group at v_i.

The index-two subgroup J_i is normal in $G_{e_i} = D_{16}$. An element of $G_{e_i} \setminus J_i$ exchanges its endpoints and interchanges the two four-groups in J_i, by the calculation above. Consequently

$$V_i \neq V_{i+1}, \qquad V_i \cap V_{i+1} = Z(J_i) = \langle z_i\rangle.$$

After the first two edges the stabilizer is J_0, of order eight. At v_1 it fixes the incoming neighbor and exchanges the other two, since its kernel there is V_1. The stabilizer after three edges is $J_0 \cap J_1 = V_1$, and the fourth edge is forced.

At v_2 the image of V_1 has order two, since $V_1 \cap V_2 = \langle z_1\rangle$. It fixes the incoming neighbor and exchanges the other two. Thus the fifth step is transitive and its stabilizer is $\langle z_1\rangle$. The sixth step is forced. At v_2, the two different Sylow subgroups J_1, J_2 have different centers, so $z_1 \neq z_2$. Since

$V_2 \cap V_3 = \langle z_2 \rangle$, the element $z_1 \in V_2$ is outside V_3. It exchanges the two continuations at v_3. This proves transitivity on all 7-arcs starting at the S_4 type.

This also gives the other starting type. Vertex transitivity on the S_4 half and the proved local transitivity imply that G is transitive on all 7-arcs starting in that half. Reversal is a G-equivariant bijection to the 7-arcs starting in the other half, since seven is odd. Hence those arcs also form a single G-orbit. Two of them with the same initial vertex can only be mapped to one another by its stabilizer, which proves the required local transitivity. At a degree-two vertex, $a_1(8) = 32 > |D_{16}| = 16$, which excludes an eighth arc. □

5 Comparison with the fixed source

The following is the complete assignment of the local arc column. It does not assert that all accompanying existence or equivalence counts in the source tables are correct; those are separate source obligations.

Source table	Rows	Exact length	Argument
1	$7, 8, 10, 11, 14, 15, 17, 18$	2	equal-type order bound
1	$1, 6, 13$	3	Proposition 3
1	$2, 3, 4$	3	Proposition 4, $s = 4$
1	$5, 9, 12, 16$	3	degree two and order bound
2	$1, 2, 3, 4, 5$	3	degree two and order bound
2	$6, 7$	2	equal-type order bound
3	$1, 9$	3	degree two and order bound
3	2	3	Proposition 3
3	$3, 4$	2	equal-type order bound
3	5	7	Proposition 7
3	6	3	Proposition 5
3	$7, 8$	4	Proposition 6
4	1	3	Proposition 4
4	2	2	equal-type order bound
4	$3, 4$	3	degree two and order bound
4	5	3	Proposition 3
5	$1, 2$	2	A_4, A_4, C_3 order bound
6	1	3	Proposition 4
6	2	3	degree two and order bound
6	3	4	Proposition 6

In Section 6 the printed second orientation of the A_5, D_{30} argument repeats types $0, 1, 0$; it must read $1, 0, 1$. In the S_4, D_{24} argument the printed "D_{18} on cosets of D_8" must read D_{24}. Proposition 5 retains the obstruction attributed there to an unnamed referee. The general arc-stabilizer method is standard and is explicitly attributed by the source to Leemans's earlier work. The contribution here is the complete uniform derivation of the table column, especially the two S_4 calculations and their common use of distinct Sylow centers.

The separate assertions on the full automorphism groups of particular graphs on source p. 386 are not settled by this note. An automorphism group containing the specified G can have greater arc transitivity; the $q = 9$ example explicitly illustrates that distinction. No upper bound on such a larger group follows merely because an outer automorphism of G interchanges two subgroup classes. No full-source completion, Lean verification, new open-question answer or compression factor is claimed by this component.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* PSL$(2, q)$*: part II*, Ars Mathematica Contemporanea **6** (2013), 365–388, Tables 1–6 and Section 6.

[2] D. Leemans, *Locally s-arc-transitive graphs related to sporadic simple groups*, Journal of Algebra **322** (2009), 882–892; attribution follows the fixed source.

Full automorphisms of the dihedral subdivision family

Human Compression Project

Abstract

Suppressing vertices of degree two reduces the (S_4, D_{16}, D_8) incidence graph to a cubic graph. Tutte's bound on cubic arc stabilizers then leaves at most an index-two extension of the specified simple group. The action on the conjugacy class of a vertex stabilizer excludes that extension. This proves the four full-automorphism assertions for this family on p. 386 of the fixed $\mathrm{PSL}(2, q)$ source, conditional only on its separately verified subgroup configurations and the classical inputs stated below.

1 Suppression and the normalizer argument

Lemma 1. *Let X be a finite connected bipartite simple graph with degrees three and two. Suppose a group G preserves types, is edge transitive and acts two-transitively on the three neighbors of each vertex of degree three. If there are more than two vertices of degree three, suppressing all vertices of degree two produces a connected simple cubic graph Y. Moreover,* $\mathrm{Aut}(X) = \mathrm{Aut}(Y)$ *under their natural actions, and G is arc transitive on Y.*

Proof. Replace each path (u, e, v), with e of degree two, by an edge uv. There are no loops because u, v are distinct neighbors in the original simple graph. A priori parallel edges are possible. At any vertex u, two of its three incident edges in the resulting multigraph may have the same other endpoint. Having the same endpoint is a G_u-invariant equivalence relation on those three edges. The two-transitive action is primitive, so if such a pair exists all three edges at u have one common endpoint v. Cubicity at v then makes these two vertices a whole connected component. This contradicts connectedness and the hypothesis on the number of vertices. Thus Y is simple and cubic; connectedness is preserved by suppression.

Every automorphism of X preserves the two degrees and hence induces an automorphism of Y. Since there are no parallel edges, a vertex of degree two is uniquely determined by its two neighbors, so the induced map is injective. Conversely an automorphism of Y extends uniquely to its subdivision by its action on edges. This proves the asserted group identification. Edge transitivity in X gives transitivity on incidences of vertices and edges in Y, which is arc transitivity. □

Lemma 2. *Let a group G act faithfully and transitively on a finite set Ω, and let its point stabilizer M satisfy* $\mathrm{N}_G(M) = M$. *If $G \triangleleft H \leq \mathrm{Sym}(\Omega)$, conjugation embeds H into* $\mathrm{Aut}(G)$. *Its image preserves the G-conjugacy class of M.*

Proof. An element c centralizing G sends a chosen point x to a point gx fixed by $M = G_x$. The two point stabilizers have the same order, so $M = G_{gx} = gMg^{-1}$ and $g \in \mathrm{N}_G(M) = M$. Therefore $cx = x$. Commutation with the transitive group now implies that c fixes every point, so $c = 1$. The kernel of conjugation from H to $\mathrm{Aut}(G)$ is thus trivial. An element $h \in H$ takes G_x to G_{hx}, and all point stabilizers are G-conjugate. This proves the last assertion. □

2 The specified family

We use the following classical inputs as premises, not as new results. For a finite connected simple cubic arc-transitive graph, Tutte's theorem gives a vertex stabilizer of order at most 48 in its full automorphism group. Indeed that full group acts regularly on the s-arcs for some $s \leq 5$, and there are $3 \cdot 2^{s-1}$ such arcs from a fixed vertex. The two parts of this result are in Tutte's 1947 and 1959 papers [2,3].

For an odd prime p in the cases below, $G = \mathrm{PSL}(2,p)$ is nonabelian simple, $\mathrm{Aut}(G) = \mathrm{PGL}(2,p)$, and the two G-conjugacy classes of subgroups isomorphic to S_4 are interchanged by every outer automorphism. These are the classical subgroup and automorphism facts used by the source; the last fact is explicitly recalled on its p. 386. The subgroups S_4 are maximal, hence self-normalizing: their normalizers could otherwise only be G, contradicting simplicity.

Theorem 3. *Let $p \in \{17, 31, 79, 97\}$. Suppose $M, L \leq G = \mathrm{PSL}(2,p)$ give one of the connected locally two-transitive coset graphs with*

$$M \cong S_4, \qquad L \cong D_{16}, \qquad M \cap L \cong D_8.$$

Then the full automorphism group of its incidence graph is exactly the given group G.

Proof. The two degrees are three and two. The half with stabilizer M has $[G : M] = p(p^2 - 1)/48 > 2$ vertices. Lemma 1 gives a connected simple cubic graph Y with the same full automorphism group H and with G arc transitive. The action of G on Y is faithful, since it is its action on G/M and the core of the proper subgroup M in the simple group G is trivial.

By Tutte's bound and orbit–stabilizer,

$$[H : G] = \frac{|H_x|}{|G_x|} \leq \frac{48}{24} = 2.$$

Thus G is normal in H, either trivially or as a subgroup of index two. Lemma 2 embeds H into $\mathrm{Aut}(G)$, with image preserving the conjugacy class of M. Every outer automorphism interchanges the two S_4 classes, so this image contains no outer automorphism. It follows that $H = G$. The equality transfers back to the incidence graph by Lemma 1. □

The proof applies to every configuration with these hypotheses; it does not inspect a graph separately for each listed prime. More generally it applies to any prime for which the same classical premises and subgroup configuration hold. The count of configurations and their fusion under outer automorphisms are separate assertions, treated by the subgroup classification. The conclusion here supplies an upper bound on the full graph group; merely observing that an outer automorphism interchanges two configurations would not have supplied that bound.

The source's cubic (S_4, S_4, D_8) graphs require a separate argument. Their specified simple group has two vertex orbits, so Tutte's bound cannot be applied to their full group without first establishing vertex transitivity. We make no such inference in this note. No whole-source completion, Lean formalization, new conjecture count or measured compression factor is claimed by this component.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* $\mathrm{PSL}(2,q)$*: part II*, Ars Mathematica Contemporanea **6** (2013), 365–388, p. 386.

[2] W. T. Tutte, *A family of cubical graphs*, Proc. Cambridge Philos. Soc. **43** (1947), 459–474, doi:10.1017/S0305004100023720.

[3] W. T. Tutte, *On the symmetry of cubic graphs*, Canadian J. Math. **11** (1959), 621–624, doi:10.4153/CJM-1959-057-2.

Pairs of symmetric parabolics and their full graph groups

Human Compression Project

Abstract

For prime $p \equiv \pm 1 \pmod 8$, the normalizer of a dihedral subgroup of order eight determines the two S_4 parabolics above it. According to one parity, these parabolics belong to the same conjugacy class or to different classes. This distinction gives the full graph groups $\mathrm{PSL}(2,p) \times C_2$ and $\mathrm{PGL}(2,p)$, respectively. The proof uses a concrete type interchange and Tutte's cubic stabilizer bound; it requires no graph automorphism search. The source's conjugacy counts are recovered, while four of its nine prime-field full-group identifications require correction.

1 The normalizer determines the parabolics

Let $p \geq 7$ be prime with $p \equiv \pm 1 \pmod 8$. Choose $\epsilon \in \{1, -1\}$ so that $p \equiv \epsilon \pmod 8$, and put

$$G = \mathrm{PSL}(2,p), \quad E = \mathrm{PGL}(2,p), \quad n = \frac{p-\epsilon}{2}, \quad d = \frac{p-\epsilon}{8}.$$

Thus $n = 4d$. We retain the classical subgroup facts used by the source: the maximal subgroups S_4 of G form two conjugacy classes, fused by E; their Sylow two-subgroups are D_8; cyclic groups of order four lie in a unique torus of order n in G and $2n$ in E; and those torus normalizers are dihedral. Also G is simple and $\mathrm{Aut}(G) = E$. The cyclic torus containing an element of order four is its projective centralizer, so its uniqueness here is literal, not just an abstract order identification.

Lemma 1. *For $B \leq G$ isomorphic to D_8,*

$$\mathrm{N}_E(B) \cong D_{16}, \qquad \mathrm{N}_G(B) = \begin{cases} D_{16}, & d \text{ even}, \\ B, & d \text{ odd}. \end{cases}$$

The subgroups B form one E-class. This class splits into two G-classes when d is even and remains one class when d is odd.

Proof. The cyclic subgroup C_4 is characteristic in B, so every normalizer of B normalizes its unique torus. Write the torus normalizer in E as

$$\langle R, S \mid R^{2n} = S^2 = 1,\ SRS = R^{-1} \rangle, \qquad \mathrm{N}_G(C_4) = \langle R^2, S \rangle,$$

choosing a reflection $S \in G$. After conjugacy in this normalizer we may take $B = \langle R^{n/2}, S \rangle$. Indeed, its reflections in G have even exponents $R^{2j}S$, and conjugation by R^j is transitive on these possibilities modulo the rotation subgroup of B. Together with conjugacy of the tori this proves that the D_8 subgroups in G form one E-class.

A rotation R^k normalizes B exactly when $(n/2) \mid 2k$, or $d \mid k$. Its normalizing rotations are therefore $\langle R^d \rangle$, of order eight. The full normalizer is $\langle R^d, S \rangle \cong D_{16}$. Since $R^k \in G$ exactly

when k is even, this normalizer lies in G when d is even. When d is odd, its intersection with G is $\langle R^{2d}, S\rangle = B$. For a subgroup of index two, an E-conjugacy class splits into two G-classes precisely when its normalizer is contained in G: this follows directly by comparing the G-orbit and E-orbit lengths. This proves the final assertion. □

Lemma 2. *Each $B \cong D_8$ is contained in exactly two subgroups M_0, M_1 isomorphic to S_4. Their intersection is B. If d is even, they belong to the same G-class and are interchanged by $\mathrm{N}_G(B)/B$. If d is odd, they belong to different G-classes and are interchanged by $\mathrm{N}_E(B)/B$. In either case there is an element t interchanging them by conjugation with $t^2 \in B$; one may take $t \in G$ exactly in the even case.*

Proof. Every S_4 contains one conjugacy class of three subgroups D_8, and $\mathrm{N}_{S_4}(D_8) = D_8$. Fix an S_4 class containing the given B. Its members containing B form a transitive $\mathrm{N}_G(B)$-set: adjust a transporter inside S_4 so that it also preserves B. The stabilizer of one member is $\mathrm{N}_G(B) \cap \mathrm{N}_G(S_4) = B$, since a maximal S_4 is self-normalizing in the simple group G. Thus the number of such members in this class is $[\mathrm{N}_G(B) : B]$.

For even d, the two B-classes are exchanged by the outer coset, as are the two S_4 classes. Each S_4 class contains only one B-class; the two S_4 classes consequently contain different B-classes. A fixed B therefore has two overgroups in one class and none in the other. Their interchange is the regular action of $\mathrm{N}_G(B)/B \cong C_2$.

For odd d, there is just one B-class. Each of the two S_4 classes supplies one overgroup. Every element of $\mathrm{N}_E(B) \setminus G$ interchanges these overgroups, because it exchanges their G-classes. In both cases the relevant normalizer is D_{16} with B of index two, so the square of an interchanging element belongs to B. Finally B is maximal in S_4; distinct equal-order overgroups containing it therefore meet exactly in B. □

These pairs give connected cubic bipartite coset graphs. The local action of S_4 on its three cosets of D_8 is S_3, and distinct maximal parabolics generate G. Thus all the rank two geometry and local two-transitivity conditions hold.

Theorem 3. *Under the source's unordered convention, the configurations with parabolics S_4, S_4 and intersection D_8 have two conjugacy classes when d is even and one when d is odd. In both cases there is one class under* $\mathrm{Aut}(G)$.

Proof. For each B the unordered pair of overgroups is unique, by Lemma 2. Its intersection recovers B. Conjugacy and automorphism classes of pairs therefore correspond exactly to the respective classes of B counted in Lemma 1. This proves the result, including the type interchange identification. □

2 A literal type interchange

Use left cosets gM_i as vertices of type i, with incidence defined by nonempty intersection. Left multiplication gives the specified type-preserving action of G. Let t be as in Lemma 2 and write $\alpha(g) = tgt^{-1}$.

Lemma 4. *The map*

$$T(gM_i) = \alpha(g)M_{1-i}$$

is a graph automorphism interchanging the two types. It satisfies $T^2 = L_{t^2}$, where L_g denotes left multiplication by $g \in G$. If $t \in G$, then $C = L_{t^{-1}}T$ is an involution centralizing the whole action of G.

Proof. Conjugation by t interchanges the two parabolics, so the displayed formula is independent of the coset representative. It carries intersecting cosets to intersecting cosets and is bijective, so it is a graph automorphism. Since $t^2 \in B \subseteq M_i$,

$$T^2(gM_i) = t^2gt^{-2}M_i = t^2gM_i = L_{t^2}(gM_i).$$

When $t \in G$, the adjusted map has the simpler expression $C(gM_i) = gt^{-1}M_{1-i}$. It commutes with every left multiplication, and its square is $gM_i \mapsto gt^{-2}M_i = gM_i$. It is nonidentity because it exchanges the two types. □

3 The full automorphism group

We retain Tutte's theorem for finite connected simple cubic arc-transitive graphs: the full vertex stabilizer has order at most 48. The source and the separate subdivision proof state this classical premise with its original references. Our graph is simple by its coset-incidence definition and connected by maximality. Lemma 4, together with the local action of G, gives arc transitivity of its full automorphism group.

Theorem 5. *The full automorphism group A of this incidence graph is*

$$A \cong \begin{cases} \mathrm{PSL}(2,p) \times C_2, & d \text{ even}, \\ \mathrm{PGL}(2,p), & d \text{ odd}. \end{cases}$$

In both cases its exact arc-transitivity length is four.

Proof. There are $2|G|/24$ vertices and A is vertex transitive. Tutte's bound gives

$$[A : G] = \frac{2|A_x|}{24} \leq 4.$$

The kernel of the action of A on its cosets of G is a normal subgroup of A contained in G, hence a normal subgroup of the simple group G. If this kernel were trivial, G would embed in S_4, impossible since $|G| \geq 168$. Therefore the kernel is G, and $G \triangleleft A$.

An element of $\mathrm{C}_A(G)$ either preserves both G-orbits of vertices or exchanges them. If it preserves both, it is the identity on each: the centralizer of a transitive coset action G/M_i is $\mathrm{N}_G(M_i)/M_i = 1$. This equality also follows directly by sending the base vertex to another vertex fixed by its stabilizer. Thus the centralizer has order at most two. If M_0, M_1 are in different G-classes, it cannot exchange the orbits, because a G-equivariant bijection of transitive G-sets requires conjugate point stabilizers.

For even d, Lemma 4 supplies precisely the nontrivial centralizer element C. Conjugation embeds $A/\langle C\rangle$ into $\mathrm{Aut}(G) = E$. Its image preserves the single G-class containing both vertex stabilizers. Every outer automorphism exchanges the two S_4 classes, so this image is $\mathrm{Inn}(G)$. Since G is centerless, it intersects $\langle C\rangle$ trivially, and consequently $A = G \times \langle C\rangle$.

For odd d, the centralizer is trivial and conjugation embeds A in E. The constructed type interchange T is outside the given type-preserving G. Therefore $|A| \geq 2|G| = |E|$, proving equality with E. This proves both full-group assertions.

In either case $|A| = 2|G|$, so $|A_x| = 24 = |G_x|$. The previously proved local length four for the specified S_4, S_4, D_8 action therefore remains the exact length for A: it has the same vertex stabilizers. The type interchange makes this a global four-arc transitive action. A fifth arc would require a vertex stabilizer divisible by $3 \cdot 2^4 = 48$, which is impossible. □

4 Comparison with the fixed source

Theorem 3 recovers Table 3, rows 7–8, including its unordered convention. Theorem 5 applies to the nine prime fields listed on printed page 386 of the fixed source, as follows.

p	d	Full graph group	Printed identification
17	2	$\mathrm{PSL}(2,17) \times C_2$	corrected
23	3	$\mathrm{PGL}(2,23)$	recovered
31	4	$\mathrm{PSL}(2,31) \times C_2$	corrected
41	5	$\mathrm{PGL}(2,41)$	recovered
47	6	$\mathrm{PSL}(2,47) \times C_2$	corrected
71	9	$\mathrm{PGL}(2,71)$	recovered
73	9	$\mathrm{PGL}(2,73)$	recovered
79	10	$\mathrm{PSL}(2,79) \times C_2$	corrected
89	11	$\mathrm{PGL}(2,89)$	recovered

The source identifies all nine full groups as $\mathrm{PGL}(2,p)$. In the four corrected cases our explicit central involution already disproves that identification: $\mathrm{PGL}(2,p)$ has trivial center, since its centralizer of the normal simple subgroup $\mathrm{PSL}(2,p)$ is trivial. The upper-bound argument proves the exact replacement, not just the failure of the printed name. There is no second principal-refutation credit: this is a surviving-claim correction in an original already registered as principally refuted. The prime-square case $q = 9$ is separate and is not covered by this proof.

The geometry family and its source counts are attributed to the original authors and the earlier work cited there. We retain the classical subgroup structure, outer action and Tutte's bound. The contribution here is the parity-controlled normalizer argument, the explicit centralizing type interchange and the resulting corrected full group theorem. The project root supplied the common Tutte/core/ centralizer upper-bound route during collaboration; the authoring agent supplied the normalizer parity, overgroup analysis and explicit type maps. The combined proof requires distinct review. It claims no whole-source completion, new open-conjecture answer, Lean verification or compression measurement.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* $\mathrm{PSL}(2,q)$*: part II*, Ars Mathematica Contemporanea **6** (2013), 365–388, Table 3 and p. 386.

[2] W. T. Tutte, *A family of cubical graphs*, Proc. Cambridge Philos. Soc. **43** (1947), 459–474.

[3] W. T. Tutte, *On the symmetry of cubic graphs*, Canadian J. Math. **11** (1959), 621–624.

The order-nine geometry from pairs and matchings

Human Compression Project

Abstract

The geometry with two S_4 parabolics in $\mathrm{PSL}(2,9)$ has a concrete model on the pairs and perfect matchings of a six-element set. This model identifies it with the symplectic generalized quadrangle of order two. An explicit duality and Tutte's stabilizer bound determine its full automorphism group and exact arc transitivity. We recover the order-nine assertions of the fixed Part II source without computing a graph automorphism group. The classical isomorphism $\mathrm{PSL}(2,9) \cong A_6$, its subgroup classes, its automorphism group, and Tutte's theorem are stated premises.

1 A concrete incidence geometry

Let $\Omega = \{1,2,3,4,5,6\}$. A *duad* is a two-element subset of Ω; a *syntheme* is a partition of Ω into three duads. Let X have the fifteen duads and the fifteen synthemes as its two vertex types. A duad is incident with a syntheme when it is one of its parts. Every vertex has degree three. The natural S_6 action is faithful, since a permutation fixing every two-element subset fixes their singleton intersections.

Lemma 1. *This incidence geometry is the symplectic generalized quadrangle of order* $(2,2)$*. Its incidence graph is connected, has girth eight and has diameter four.*

Proof. For a duad d not in a syntheme L, its two letters occur in two different parts of L. Exactly one part e of L is disjoint from d. There is a unique syntheme containing d and e: its third part consists of the two unused letters. Thus there is a unique point of L collinear with d. Two duads are on at most one syntheme, and two synthemes share at most one duad. These facts are precisely the generalized quadrangle axioms, with three incidences at each vertex.

For completeness, its symplectic realization is equally explicit. Let $W \subset \mathbb{F}_2^6$ consist of the vectors with even coordinate sum, and put $V = W/\langle(1,1,1,1,1,1)\rangle$. The ordinary dot product restricts to an alternating form on W; its radical is the displayed one-dimensional subspace. It therefore induces a nondegenerate alternating form on the four-dimensional space V. Every nonzero class has a unique representative of weight two, so its fifteen projective points are precisely the duads. Distinct duads are orthogonal exactly when they are disjoint. The sum of two disjoint duads is represented by their complementary duad, since adding the all-ones vector replaces a set by its complement. Consequently the totally isotropic two-dimensional subspaces are exactly the synthemes. This proves the claimed realization.

The quadrangle axiom gives distance three for a nonincident point and line, and distance at most four between any two vertices. Two intersecting duads are not collinear, so their distance is four. There is no four-cycle because two points have at most one common line, and no six-cycle by the uniqueness in the quadrangle axiom. There is an eight-cycle: the successive duads $12, 34, 15, 36$ are joined by the four synthemes $12|34|56$, $15|26|34$, $15|24|36$, $12|36|45$. Thus the diameter and girth are exactly as asserted. □

Here ab denotes $\{a, b\}$. The dot-product construction also identifies the type-preserving S_6 action with $\mathrm{Sp}_4(2)$. Indeed it embeds S_6 faithfully in that group, whose order is $15 \cdot 8 \cdot 3 \cdot 2 = 720$: choose the first nonzero symplectic basis vector, its partner, and then a symplectic basis of their two-dimensional orthogonal complement.

2 The specified simple group and its configuration

Put $G = A_6$. We use the classical isomorphism $G \cong \mathrm{PSL}(2, 9)$ and the two maximal S_4 classes in this group. These are precisely the two classes supplied by the subfield $\mathrm{PGL}(2, 3)$ case of the Dickson subgroup theorem; they are not conclusions of the geometry classification being audited.

Lemma 2. *The duad and syntheme stabilizers in G represent those two S_4 classes. An incident pair has intersection D_8. The action on each local set of three neighbors is two-transitive. Every configuration of two distinct S_4 subgroups meeting in D_8 is conjugate to one such incident pair, under the unordered convention.*

Proof. The duad stabilizer is $(S_2 \times S_4) \cap A_6$, isomorphic to S_4 by its action on the other four letters: parity uniquely determines the action on the duad. The syntheme stabilizer is the even part of $C_2 \wr S_3$, namely $C_2^2 \rtimes S_3 \cong S_4$. It acts transitively on the six letters, whereas the duad stabilizer has orbits of lengths two and four. They cannot be conjugate even in S_6, and hence represent the two given G-classes. Both classes have fifteen members, since both actions are transitive.

The duad stabilizer acts as the full S_3 on the three matchings of the remaining four letters; the syntheme stabilizer acts as the full S_3 on its three parts. Thus both local actions are two-transitive. An incident-pair stabilizer has order $24/3 = 8$ and is a Sylow two-subgroup of S_4, hence D_8. Denote it by B.

Its orbits on the six letters have sizes two and four. A normalizer of B in A_6 must preserve the orbit of size two and therefore belongs to the duad stabilizer. A Sylow two-subgroup of S_4 is self-normalizing, so $N_G(B) = B$. There are therefore $360/8 = 45$ conjugate Sylow two-subgroups of G. Each S_4 contains three of them. For each of the two S_4 classes, counting incidences gives $15 \cdot 3 = 45$; transitivity on the Sylow subgroups shows that each one belongs to exactly one S_4 of each class. It follows that every eligible pair is conjugate to the displayed pair. Its coset incidence graph is X: the natural transitive actions identify both coset sets, and flag transitivity identifies their incidence relation. □

3 An explicit duality and the full group

The following table gives a permutation d of the thirty vertices. Each row is a transposition between the duad and the syntheme shown. In particular $d^2 = 1$ and d interchanges the two types.

Duad	Syntheme	Duad	Syntheme	Duad	Syntheme
12	12|34|56	13	14|25|36	14	13|26|45
15	16|24|35	16	15|23|46	23	16|23|45
24	15|24|36	25	13|25|46	26	14|26|35
34	12|35|46	35	15|26|34	36	13|24|56
45	14|23|56	46	16|25|34	56	12|36|45

For every incidence $a \in L$, the syntheme $d(a)$ contains the duad $d(L)$. This follows by substituting the three parts of each of the fifteen synthemes in the table. Thus d is a graph automorphism. The accompanying literal reader checks exactly these forty-five incidences, the permutation property and the reversal of types. The witness was found by a graph-isomorphism routine; its verification uses neither that routine nor a group-order computation.

Theorem 3. *The full automorphism group of X is isomorphic to* $\mathrm{P\Gamma L}(2,9)$ *and has order* 1440*. It is five-arc-transitive and is not six-arc-transitive.*

Proof. Let $H = \mathrm{Aut}(X)$. It contains the faithful type-preserving S_6 and the type-reversing d, so $|H| \geq 2 \cdot 720 = 1440$. The action is vertex transitive; the local actions in Lemma 2 make it arc transitive. Tutte's theorem for finite connected simple cubic arc-transitive graphs gives $|H_x| \leq 48$. Since X has thirty vertices, $|H| \leq 30 \cdot 48 = 1440$. Equality holds.

To identify the group, its subgroup $G = A_6$ has index four. The action of G on the four cosets H/G is a homomorphism to S_4. Simplicity and $|G| = 360 > 24$ force this homomorphism to be trivial. Hence G lies in its core in H and is normal in H. The two G-orbits on vertices have nonconjugate stabilizers by Lemma 2, so a permutation centralizing G cannot interchange them. Within either orbit, its centralizer is $N_G(M)/M = 1$, since the maximal S_4 stabilizer is self-normalizing in the simple group. Thus $C_H(G) = 1$, and conjugation embeds H in $\mathrm{Aut}(G)$. The classical automorphism theorem identifies $\mathrm{Aut}(\mathrm{PSL}(2,9))$ with $\mathrm{P\Gamma L}(2,9)$, also of order 1440; the embedding is therefore an isomorphism.

Tutte's theorem further says that the full group is regular on s-arcs for some $s \leq 5$. The number from a fixed vertex is $3 \cdot 2^{s-1}$. Since $|H_x| = 48$, necessarily $s = 5$. At length six there are 96 such arcs, exceeding the stabilizer order, so six-arc transitivity is impossible. □

4 Scope and attribution

This proves the $q = 9$ row with two S_4 parabolics in Table 6 and the generalized-quadrangle and full-group assertions on p. 386 of the fixed Part II source. The specified G-action has local arc length four by the separately accepted stabilizer argument; the full group has length five as proved here. The classical duad–syntheme model, symplectic realization and Tutte results are reused, not claimed as new discoveries. Our contribution here is the complete derivation and explicit witness within this source audit. The other source families remain separate; this component adds no whole-original completion, new conjecture answer, Lean verification or compression measurement.

References

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